

Lifetime Consumption-Portfolio Choice under Trading Constraints, Recursive Preferences, and Non-Tradeable Income*

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Abstract

We consider the lifetime consumption-portfolio problem in a competitive securities market with essentially arbitrary continuous price dynamics, a possibly nontradeable income stream, and convex constraints on the vector of market values of financial positions. (The setting extends Schroder and Skiadas, 2002, where the endowment is assumed tradeable and constraints are imposed in terms of wealth proportions.) For any utility function with a supergradient density, we develop the first-order conditions of optimality, a side-product being the characterization of a constrained notion of state-pricing. The methodology is applied to generalized continuous-time recursive utility, allowing for first and second-order risk-aversion that can depend on the risk source, reflecting the source's "ambiguity." Within this class, we isolate a more tractable formulation in which preferences exhibit no wealth effects (an example being time-additive expected discounted exponential utility), and there is unrestricted trading in a money market and a suitably defined consol bond. In this case, we derive closed-form solutions for the optimal consumption and trading strategy in terms of the solution to a single constrained backward stochastic differential equation (BSDE), which in a Markovian setting maps to a PDE. Methodologically, we develop the utility gradient approach, but for the wealth-invariant case we also verify the solution using the dynamic programming approach, without having to assume a Markovian structure. Finally, we present a class of parametric examples in which the PDE characterizing the solution simplifies to a system of ordinary differential equations (of the Riccati type).

1 Introduction

This paper analyzes the lifetime consumption-portfolio problem for an agent who can trade in a competitive securities market with essentially arbitrary continuous price dynamics, under the constraint that the vector of investment values must lie in a given convex set at all times. The setting extends Schroder and Skiadas (2002b), hereafter abbreviated to SS, by allowing nontradeable income and constraints on the vector of portfolio values as opposed to wealth proportions.

For any utility function with a well-defined gradient density (or supergradient density in the non-smooth case), we develop the first-order conditions of optimality, stating that the utility gradient density is a state-price density at the candidate optimum. The definition of a state-price density (or pricing kernel) in this context extends the usual unconstrained notion, essentially stating that every feasible incremental consumption plan is priced non-negatively. A contribution unrelated to preferences is the characterization of state prices in this sense. Defining a quasi-arbitrage as a feasible incremental trade that results in a positive incremental cash flow, a state-price density corresponds to a subjective valuation rule that is consistent with the absence of a quasi-arbitrage. Given monotone preferences, optimality rules out quasi-arbitrage, while the utility gradient at the optimum selects one of many possible valuation rules consistent with no quasi-arbitrage.

The first-order conditions are applied to generalized recursive utility, a utility specification proposed by Lazrak and Quenez (2002) as a way of nesting the recursive utility of Duffie and Epstein (1992) (a special case of which is continuous-time Epstein and Zin (1989) utility, as well as time-additive expected utility), the multiple-prior formulation of Chen and Epstein (2002), and, as shown in Skiadas (2002), the robust-control type criteria used by Anderson, Hansen, and Sargent (2000) and Maenhout (1999). As in SS, the utility specification allows for first and second-order risk aversion (in the sense of Segal and Spivak (1990)) that can be dependent on the source of risk, a dependence that can be thought of as reflecting the source's ambiguity or uncertainty in the sense of Ellsberg's experiments. We henceforth refer to generalized recursive utility simply as recursive utility, except for emphasis.

The resulting first-order conditions of optimality are expressed as a coupled system of forward and backward recursive equations. The forward recursion applies to wealth, merely keeping track of the budget constraint, starting with the agent's initial wealth. (While the general theory does not require a Markovian structure, in a Markovian implementation one would also add to the set of forward recursions the state dynamics.) The backward recursions compute utility and corresponding Lagrange multiplier, or shadow price, of wealth, starting with the utility and marginal utility, respectively, for terminal wealth, and continuing backward on the information tree. Significant simplifications in the solution result under two types of assumptions: scale-invariance or translation-invariance. In both cases, there is a simple relationship between the forward and backward components which uncouples the forward-backward recursion of the first-order conditions.

The main ingredients of the scale-invariant formulation are homothetic preferences (such as time-additive HARA, Epstein-Zin recursive utility, or multiple-prior versions), a traded endowment, and constraints that are imposed on portfolio positions expressed as proportions of wealth (including short-sale constraints and missing markets). As a result of these assumptions, if the agent's initial financial wealth is scaled by some factor, then the agent's optimal investment values and consumption level is scaled by the same factor. In certainty equivalent form, the optimal utility process is simply the product of wealth and the shadow price of wealth, and the backward recursion of the latter is uncoupled from the wealth dynamics. The scale-invariant case is developed in detail in SS.

Besides the general formulation, this paper focuses on a translation-invariant formulation, in which the income stream can be nontradeable. Preferences are assumed to be translation-invariant relative to some numeraire plan γ , meaning that if consumption plan a is preferred to consumption plan b , then $a + k\gamma$ is preferred to $b + k\gamma$ for any choice of a scalar k . Examples are preferences represented by time-additive expected discounted exponential utility, as well as recursive and multiple-prior extensions with first or second-order (possibly source-dependent) risk aversion. The translation-invariant formulation allows general constraints on the values invested in risky assets, but assumes unrestricted trading in a default-free money market as well as a "consol bond" (or portfolio) that generates γ as a dividend stream. In this case, taking the market-value of the consol bond as the unit of account, the shadow price of wealth is always one, while the optimal utility process (in certainty equivalent form) is equal to the wealth process plus a process satisfying a backward recursion that does not depend on wealth.

As in SS, methodologically we extend to the constrained case the utility gradient approach, originating in Cox and Huang (1989) and Karatzas, Lehoczky, and Shreve (1987) for the case of additive utility and complete markets, and extended in Duffie and Skiadas (1994), Schroder and Skiadas (1999), and El Karoui, Peng, and Quenez (2001) to recursive utility settings. The utility gradient approach is essentially the statement of the infinite-dimensional Kuhn-Tucker conditions for the constrained problem. A related literature on constrained portfolio selection with additive utilities (He and Pearson (1991), Karatzas, Lehoczky, Shreve, and Xu (1991), Shreve and Xu (1992), Cvitanić and Karatzas (1992), Kramkov and Schachermayer (1999), Hugonnier and Kramkov (2002)) focuses on dual formulations that involve the minimization of the value function over all possible completions of the market. Cuoco (1997) showed existence of an optimum under non-traded income and additive utility.

An alternative to the utility gradient approach, dating back to the seminal work of Merton (1971), is dynamic programming, which in this context is sometimes incorrectly perceived as applying only with an underlying set of Markov states. For the wealth-invariant case, an appendix gives a dynamic programming optimality verification argument without the need for a Markovian framework. For the scale-invariant case, an analogous non-Markovian argument is given in SS. The dynamic-programming approach focuses on the indirect utility for wealth, while the utility gradient approach focuses on the direct utility for con-

sumption. The two are related by an envelope-type condition at the optimum, and complement each other in our understanding of the optimal demand picture. Related Markovian dynamic programming treatments with i.i.d. instantaneous returns and time-additive expected utility include Duffie, Fleming, Soner, and Zariphopoulou (1997), who considered HARA utility and non-tradeable income, and Zariphopoulou (1994) who included no income stream, but allowed for possibly binding borrowing and bankruptcy constraints.

Whatever the methodology leading to them, the optimality conditions take the form of a constrained forward-backward recursion, also known as a Forward-Backward Stochastic Differential Equation (FBSDE). In a Markovian setting, the FBSDE can be reduced to a PDE system. While numerical PDE methods can be used, there is a particularly tractable class of preferences and dynamics that reduces the PDE system to a system of ordinary differential equations (Riccati equations). In the scale-invariant case (developed in SS), this class corresponds to a Lagrange multiplier for wealth that is exponential-quadratic in the Markov states. In the wealth-invariant case, considered in the last section of this paper, the value function can be expressed as wealth plus a quadratic of the state vector. The corresponding price dynamics are essentially those specified in Liu (2001), who extended the specification in the last section of Schroder and Skiadas (1999). Mathematically, these results are related to exponential-affine solutions in the term-structure literature (see Duffie and Kan (1996), Duffie, Filipovic, and Schachermayer (2001), and the recent survey by Piazzesi (2002)).

As noted earlier, the general characterization of the optimum applies to any utility with a well-defined (super)gradient density. While in this paper we only pursue applications with forward-looking utilities, the utility gradient computations in Duffie and Skiadas (1994) show that the method can also be applied to utilities with path dependence, such as habit-formation (as in Detemple and Zapatero (1991) and Schroder and Skiadas (2002a)).

The remainder of the paper is organized in seven sections and two appendices. In Section 2 the market is defined, and optimality is characterized in terms of the relationship between a utility (super)gradient density, and a constrained notion of a state-price density. The latter is characterized in Section 3, and is related to the notion of a quasi-arbitrage. In Section 4 we apply the characterization of optimality to generalized recursive utility, with Section 5 specializing to the translation-invariant case. Section 6 specializes further to the important “quasi-quadratic” specification, which includes all wealth-invariant Duffie and Epstein (1992) utilities (for example, additive exponential), but is also broad enough to allow for first or second-order risk aversion that can depend on the risk source. Section 7 contains more detailed non-Markovian solutions for the quasi-quadratic case, emphasizing the effect of first-order risk aversion, and the effect of a non-tradeable endowment. Section 8 contains incomplete-markets Markovian applications, including a specification for which the first-order conditions reduce to a system of Riccati equations. Appendix A gives a dynamic programming verification argument for the wealth-invariant case, while Appendix B contains proofs.

2 Market and Optimality

This section extends the setting of Schroder and Skiadas (2002b), abbreviated to SS throughout the paper, to allow for an income stream and constraints in the value of portfolio positions, as opposed to constraints on portfolio weights.

Uncertainty is modeled by a probability space $(\Omega, \mathcal{F}, P)$ supporting a d -dimensional standard Brownian motion B over the finite time horizon $[0, T]$. All processes appearing in this paper are assumed to be progressively measurable with respect to the (augmented) filtration $\{\mathcal{F}_t : t \in [0, T]\}$ generated by B . We also assume that $\mathcal{F}_T = \mathcal{F}$. The conditional expectation operator $E[\cdot | \mathcal{F}_t]$ will be abbreviated to E_t throughout. The qualification “almost surely” (or a.s.) will be often omitted where it is implied by the context.

Given any subset S of a Euclidean space, $\mathcal{L}(S)$ denotes the set of all S -valued (progressively measurable) processes. We will also make frequent use of the following spaces, where $\|\cdot\|$ denotes the usual Euclidean norm:

$$\begin{aligned}\mathcal{L}_p(S) &= \left\{ x \in \mathcal{L}(S) : \int_0^T \|x_t\|^p dt < \infty \text{ a.s.} \right\}, \quad p \in \{1, 2\}, \\ \mathcal{S}_p &= \left\{ x \in \mathcal{L}(\mathbb{R}) : E \left[\left(\text{ess sup}_{t \in [0, T]} |x_t|^p \right) \right] < \infty \right\}, \\ \mathcal{H} &= \left\{ x \in \mathcal{L}(\mathbb{R}) : E \left[\int_0^T x_t^2 dt + x_T^2 \right] < \infty \right\}.\end{aligned}$$

The space $\mathcal{H}$ is a Hilbert space under the inner product

$$(x|y) = E \left[\int_0^T x_t y_t dt + x_T y_T \right].$$

As usual, we identify any elements x and $\tilde{x}$ in $\mathcal{H}$ such that $(x - \tilde{x}|x - \tilde{x}) = 0$. Statements of the form “ $x_t = y_t$ ” or “ $x_t \in S$ ” or “ x is valued in S ” should be interpreted to mean that the indicator function of the condition is equal to one as an element of $\mathcal{H}$.

There is a securities market that allows instantaneous default-free borrowing and lending at a rate given by the process r such that $\int_0^T |r_t| dt < \infty$ a.s. We refer to trading at this rate as the “money market.” The rest of the securities market consists of trading in $m \leq d$ risky assets, whose instantaneous excess returns (relative to r) are represented by the m -dimensional Ito process R , with dynamics

$$dR_t = \mu_t^R dt + \sigma_t^{R'} dB_t,$$

where $\mu^R \in \mathcal{L}_1(\mathbb{R}^m)$ and $\sigma^R \in \mathcal{L}_2(\mathbb{R}^{d \times m})$. We assume throughout that σ_t^R is everywhere full-rank for all t (and therefore is invertible if $m = d$).

A *trading plan* is any process $\phi \in \mathcal{L}(\mathbb{R}^m)$ such that

$$\int_0^t |\phi_s' \mu_s^R| + \phi_s' \sigma_s^{R'} \sigma_s^R \phi_s ds < \infty \quad \text{a.s. for all } t < T.$$

(All vectors are assumed to be column vectors, with a prime denoting transposition.) We interpret the i^{th} component of ϕ_t as the time- t market value of the agent's investment in security $i \in \{1, \dots, m\}$, the remaining wealth being invested in the money market. A (financial) *wealth process* is any process W such that $W^- \in \mathcal{S}_2$, where $W_t^- = \max\{0, -W_t\}$. (This restriction is imposed to rule out doubling-type strategies.) A *plan* (c, ϕ, W) is a triple of a consumption plan c , a trading plan ϕ , and a wealth process W .

We consider an agent characterized by the primitives $(\mathcal{C}, U_0, w_0, e, S)$. The set $\mathcal{C} \subseteq \mathcal{H}$ is convex, and its elements are the *consumption plans*. Given any consumption plan $c \in \mathcal{C}$, we think of c_t , $t < T$, as the time- t consumption rate and c_T as a terminal lump-sum consumption or bequest. The function $U_0 : \mathcal{C} \rightarrow \mathbb{R}$ is an increasing utility function over consumption plans. The positive scalar $w_0 > 0$ is an initial financial wealth, while $e \in \mathcal{H}$ is an endowed income stream, with e_t representing a time- t income rate, and e_T representing a terminal lump-sum payment. The set S , a (non-empty) convex subset of $\mathbb{R}^{m+1}$, is the agent's *constraint set*, in which the agent's vector of risky-asset investment values and total wealth is restricted to lie at all times.

The plan (c, ϕ, W) is *feasible* (for the agent) if

$$(\phi_t, W_t) \in S \quad \text{a.s. for all } t,$$

and it satisfies the *budget equation*:

$$W_0 = w, \quad dW_t = (W_t r_t + e_t - c_t) dt + \phi_t' dR_t, \quad c_T = W_T + e_T. \quad (1)$$

Remark 1 (a) *Given a general constraint set, it is possible that some free disposal of wealth is optimal (see Cuoco (1997)). While the paper's arguments extend readily if a free-disposal term is included in the wealth dynamics, the issue does not arise in our main applications, and therefore free-disposal is omitted for simplicity of exposition.*

(b) *The analysis goes through with a constraint set S that is time and state dependent, meaning that S can be a mapping from $\Omega \times [0, T]$ to the set of subsets of $\mathbb{R}^{m+1}$ (satisfying some technical measurability assumption).*

A consumption plan c is *feasible* if (c, ϕ, W) is feasible for some trading plan ϕ and wealth process W . A consumption plan c is *optimal* if it is feasible and $U(c) \geq U(\tilde{c})$ for any other feasible consumption plan $\tilde{c}$, while the plan (c, ϕ, W) is *optimal* if it is feasible and c is optimal. A trading plan ϕ is *feasible* (resp. *optimal*) if (c, ϕ, W) is feasible (resp. optimal) for some (c, W) .

Fixing a reference plan (c, ϕ, W) , we formulate conditions for its optimality, in terms of a utility (super)gradient density. The abstract argument below is the same as in SS, applied to the current more general context.

A process $x \in \mathcal{H}$ is a *feasible incremental cash flow* at c if $c + x$ is a feasible consumption plan. A process π is a *state-price density* at c if $\pi \in \mathcal{H}$ and $(\pi|x) \leq 0$ for every feasible incremental cash flow x at c .

The process $\pi \in \mathcal{H}$ is a *utility supergradient density* of U_0 at c if

$$U_0(c + x) \leq U_0(c) + (\pi|x) \quad \text{for all } x \text{ such that } c + x \in \mathcal{C}. \quad (2)$$

The process $\pi \in \mathcal{H}$ is a *utility gradient density* of U_0 at c if

$$(\pi|x) = \lim_{\alpha \downarrow 0} \frac{U_0(c + \alpha x) - U_0(c)}{\alpha}, \text{ for all } x \text{ such that } c + \alpha x \in \mathcal{C} \text{ for some } \alpha > 0.$$

If π is a supergradient density of U_0 at c and the utility gradient of U_0 at c exists, then the utility gradient density equals π .

Proposition 2 *Suppose (c, ϕ, W) is a feasible plan. If $\pi \in \mathcal{H}$ is both a supergradient density of U_0 at c and a state-price density at c , then the plan (c, ϕ, W) is optimal. Conversely, if the plan (c, ϕ, W) is optimal and $\pi \in \mathcal{H}$ is a utility gradient density of U_0 at c , then π is a state-price density at c .*

Proof. If $c + x$ is a feasible consumption plan and π is a state price density, then $(\pi|x) \leq 0$. If π is also a supergradient density, then inequality (2) implies $U_0(c + x) \leq U_0(c)$, proving optimality of c . Conversely, suppose c is optimal and π is a utility gradient density of U_0 at c . Let x be any feasible incremental cash flow at c . Since S and $\mathcal{C}$ are convex, $c + \alpha x$ is a feasible consumption plan for all $\alpha \in [0, 1]$. Maximization with respect to α at zero implies that $(\pi|x) \leq 0$, proving the state-price density property of π . ■

To apply this proposition, we need to characterize state-price densities, and to compute utility (super)gradient densities given a utility specification. These are the respective topics of the following two sections.

3 Characterization of State Prices

Fixing the reference plan (c, ϕ, W) , in this section we formulate sufficient and, under some assumptions, necessary conditions for a strictly positive Ito process $\pi \in \mathcal{H}$ to be a state-price density at c . Given Proposition 2, if π is a utility (super)gradient density at c , then this section's conditions become the first-order conditions for optimality of the reference plan (c, ϕ, W) . Other than this interpretation, however, preferences play no role in this section's results.

Given a strictly positive Ito process π with dynamics

$$\frac{d\pi_t}{\pi_t} = -(r_t + \zeta_t) dt - \eta'_t dB_t, \quad (3)$$

for some processes $\zeta \in \mathcal{L}(\mathbb{R})$ and $\eta \in \mathcal{L}(\mathbb{R}^d)$, we are interested in characterizing the state-price density property of π in terms of the coefficients (ζ, η) . For this purpose we define the support function $\delta : \mathbb{R}^{m+1} \rightarrow (-\infty, \infty]$ by

$$\delta(\varepsilon, \zeta) = \sup \{v'\varepsilon - w\zeta : (v, w) \in S\}. \quad (4)$$

Given a feasible plan (c, ϕ, W) , the key condition for π to be a state-price density at c is

Condition 3 $\phi'_t \varepsilon_t - W_t \zeta_t = \delta(\varepsilon_t, \zeta_t)$, where $\varepsilon_t = \mu_t^R - \sigma_t^{R'} \eta_t$.

Although the above condition is purely a state-price density characterization, it has an intuitive interpretation in a context in which (c, ϕ, W) is an optimal consumption plan and π is the utility gradient density at c . In this case we can think of π as being the agent's subjective marginal pricing kernel at the optimum, which may not be consistent with the way the market prices assets, due to the agent's constraints. (For example, in equilibrium, an agent can regard a stock overvalued if the agent is constrained from short-selling the stock.) If π follows the dynamics of equation (3), the agent assesses a subjective short-rate $r + \zeta$ and a subjective market-price-of-risk process η . Given a portfolio-wealth pair $(v, w) \in S$, $v'\varepsilon - w\zeta$ can be thought of as the instantaneous expected mispricing benefit to the agent, which Condition 3 states must be maximized along the optimal path. Further interpretations of the processes (ζ, η) are provided after the statement of the section's main results below.

Example 4 (scale-invariant constraints) *In SS we analyze the special case in which $e = 0$ (or, equivalently, e can be sold for some initial wealth), consumption and wealth are always strictly positive, and the set S is invariant to scaling of wealth, that is, $(v, w) \in S \iff (v/w, 1) \in S$. In this case, $\delta(\varepsilon, \zeta) < \infty$ implies that $\delta(\varepsilon, \zeta) = 0$ and $\zeta = \zeta(\varepsilon) = \sup \{\psi'\varepsilon : (\psi, 1) \in S\}$. Condition 3 becomes $\psi'_t \varepsilon_t = \zeta_t(\varepsilon_t)$, where $\psi_t = \phi_t/W_t$ is the vector of risky asset portfolio weights. (This is Condition 4 in SS.)*

Example 5 (wealth-invariant constraints) *Later in this paper we explore applications with a possibly non-traded endowment e , and a constraint set of the form $S = \{(v, w) : v \in \Phi\}$, for some convex set $\Phi \subseteq \mathbb{R}^m$. In this case, $\delta(\varepsilon, \zeta) < \infty$ implies $\zeta = 0$, and Condition 3 becomes $\phi'_t \varepsilon_t = \delta(\varepsilon_t, 0)$.*

The sufficiency of Condition 3 for π to be a state-price density holds quite generally:

Proposition 6 *Suppose (c, ϕ, W) is a feasible plan, the strictly positive process $\pi \in \mathcal{S}_2$ follows the dynamics (3) for some $\zeta \in \mathcal{L}(\mathbb{R})$ and $\eta \in \mathcal{L}(\mathbb{R}^d)$ satisfying Condition 3, and $\pi W \in \mathcal{S}_1$. Then π is a state-price density at c .*

The necessity of Condition 3 for π to be a state-price density requires some more structure. Consider first the following lemma:

Lemma 7 *Suppose (c, ϕ, W) is a feasible plan, and the strictly positive process $\pi \in \mathcal{H}$ follows the dynamics (3). Given any feasible plan of the form $(c + x, \phi + y, W + z)$, suppose that $\Delta = y'\varepsilon - z\zeta \in \mathcal{H}$ and $\pi z \in \mathcal{S}_1$. Then $(\pi|x) = (\pi|\Delta)$.*

The basic idea behind a (partial) converse to the last Proposition is to show that the violation of Condition 3 leads to the application of Lemma 7 with $0 \neq \Delta \geq 0$. Such an argument can be made quite generally in a setting in which cumulative consumption is allowed to jump. In settings in which cumulative consumption is required to be absolutely continuous, some sort of technical

limiting argument is required, whose details depend on the type of application. For example, in the scale-invariant context of Example 4, a converse to Proposition 6 is proved in SS. This paper's main applications are in the context of Example 5, in which case the following converse applies:

Proposition 8 *Suppose that $S = \{(v, w) : v \in \Phi\}$ for some convex set $\Phi \subseteq \mathbb{R}^m$, that $\mathcal{C}$ is a convex set such that $c + b \in \mathcal{C}$ for any $c \in \mathcal{C}$ and bounded $b \in \mathcal{L}(\mathbb{R})$, and that r is bounded. Suppose further that (c, ϕ, W) is a feasible plan, and the strictly positive process $\pi \in \mathcal{S}_1$ has dynamics (3) for some $(\zeta, \eta) \in \mathcal{L}(\mathbb{R}) \times \mathcal{L}(\mathbb{R}^d)$. If π is a state-price density at c , then Condition 3 holds with $\zeta = 0$.*

Remark 9 *The assumption that r is bounded is made for simplicity. It can be relaxed by taking the value of one unit initially invested in the money market as the numeraire, and expressing the above assumptions in terms of the new unit of account.*

To further interpret the pair (ζ, η) , we define a *quasi-arbitrage* at c as any non-zero positive feasible incremental cash flow at c . If (c, ϕ, W) is optimal and U_0 is strictly increasing, then it is necessary that there exists no quasi-arbitrage at c . The non-existence of a quasi-arbitrage implies that the time- t portfolio maximizes instantaneous expected return given the time- t (financial) wealth W_t and the instantaneous volatility matrix $\sigma_t^R \phi_t$. In this maximization problem, the random variables $r_t + \zeta_t$ and η_t represent the shadow prices of wealth and volatility, respectively. More formally, we define the function $\mathcal{R}_t : \Omega \times \mathbb{R}^{1+d} \rightarrow (-\infty, +\infty]$ that assigns to $(w, \sigma) \in \mathbb{R}^{1+d}$ the random variable

$$\mathcal{R}_t(w, \sigma) = \text{ess sup} \{wr_t + v'\mu_t^R : v \text{ satisfies } (v, w) \in S \text{ and } \sigma_t^R v = \sigma\}. \quad (5)$$

Non-existence of a quasi-arbitrage implies the condition

$$\mathcal{R}_t(W_t, \sigma_t^R \phi_t) = W_t r_t + \phi_t' \mu_t^R. \quad (6)$$

Let $\partial \mathcal{R}_t(w, \sigma)$ denote the (state-by-state) *superdifferential* of $\mathcal{R}_t$, that is, the set of all (a, b) such that $\mathcal{R}_t(w + h, \sigma + z) \leq \mathcal{R}_t(w, \sigma) + ah + b'z$ for all (h, z) .

Proposition 10 *Given identity (6), Condition 3 is equivalent to*

$$(r_t + \zeta_t, \eta_t) \in \partial \mathcal{R}_t(W_t, \sigma_t^R \phi_t).$$

Proof. Given any $(v, w) \in S$, $(r_t + \zeta_t, \eta_t) \in \partial \mathcal{R}_t(W_t, \sigma_t^R \phi_t)$ implies

$$\mathcal{R}_t(w, \sigma_t^R v) - \mathcal{R}_t(W_t, \sigma_t^R \phi_t) \leq (r_t + \zeta_t)(w - W_t) + \eta_t'(\sigma_t^R v - \sigma_t^R \phi_t).$$

Using the fact that $\mathcal{R}_t(w, \sigma_t^R v) \geq r_t w + v'\mu_t^R$ and equality (6), it follows that

$$-w\zeta_t + v\varepsilon_t \leq -W_t\zeta_t + \phi_t'\varepsilon_t,$$

where $\varepsilon_t = \mu_t^R - \sigma_t^{R'} \eta_t$. Taking the supremum over $(w, v) \in S$, results in Condition 3. The converse is shown similarly, by reversing the above steps. ■

Remark 11 *A corollary of the above discussion (via a measurable-selections argument) is that the lack of quasi-arbitrage implies the existence of processes (ζ, η) satisfying Condition 3. This generalizes a familiar argument linking the lack of arbitrage opportunities without constraints to the existence of a market-price-of-risk process (see Theorem 4.2 of Karatzas and Shreve (1998), or Lemma 6G of Duffie (2001)).*

Yet another interpretation of the coefficients (ζ, η) is obtained by assuming that $m = d$ and σ_t^R is invertible. This assumption entails no loss in generality, since impossibility of trading in a security can be modeled through S (as in Remark 1 of SS). In this case, a state-price density π with dynamics (3) is the unique, up to positive scaling, state-price density in a fictitious unconstrained complete market obtained from the original market by modifying the short rate process from r to $\hat{r} = r + \zeta$, and the instantaneous expected excess returns process from μ^R to $\hat{\mu}^R = \sigma^{R'}\eta$. Given Condition 3, feasibility of a plan in the original market and the fictitious complete market are equivalent. It follows that if π is a utility supergradient density at c satisfying Condition 3, then the plan (c, ϕ, W) is optimal in the original market given the constraint $(\phi_t, W_t) \in S$, and is also optimal in the unconstrained fictitious complete market. In settings, such as that of Proposition 8 or Proposition 3 of SS, in which Condition 3 is necessary for optimality of (c, ϕ, W) , one concludes that optimality of (c, ϕ, W) implies optimality of the same plan in the above-defined fictitious unconstrained complete market.

4 Generalized Recursive Utility

In this section we apply the optimality conditions discussed so far to the case of (generalized) recursive utility. We begin with the utility specification, followed by the formulation of the first-order conditions of optimality.

Consumption values are restricted to lie in an open interval $C \subseteq \mathbb{R}$, typically $C = (0, \infty)$ or $C = \mathbb{R}$, meaning that every $c \in C$ is of the form $c : \Omega \times [0, T] \rightarrow C$. Utility is specified in terms of a (progressively measurable) function $F : \Omega \times [0, T] \times C \times \mathbb{R}^{1+d} \rightarrow \mathbb{R}$, called an *aggregator*, such that the random variable $F(T, c, U, \Sigma)$ does not depend on the arguments (U, Σ) , which are therefore notationally suppressed. Also fixed is a subset $\mathcal{U}$ of $\mathcal{S}_2$ to be thought of as the set of all possible utility processes. For the rest of this paper, we assume:

Condition 12 (Standing Assumption) *Given any consumption plan c , the utility value $U_0(c)$ is determined as the initial value of the unique process $U = U(c)$ in $\mathcal{U}$ (a subset of $\mathcal{S}_2$) that solves the BSDE*

$$dU_t = -F(t, c_t, U_t, \Sigma_t^U) dt + \Sigma_t^{U'} dB_t, \quad U_T = F(T, c_T). \quad (7)$$

Moreover, $F(\omega, t, \cdot)$ is a concave function for all (ω, t) , and the derivative, $F_c(\omega, t, \cdot, U, \Sigma)$, of $F(\omega, t, \cdot, U, \Sigma)$ exists and maps C onto $(0, \infty)$.

The aggregator F need not be smooth in (U, Σ) , a generality that allows for first-order risk aversion. The *superdifferential* of F with respect to the variables (U, Σ) at $(\omega, t, c, U, \Sigma)$ is defined as the set $(\partial_{U, \Sigma} F)(\omega, t, c, U, \Sigma)$ of all pairs $(a, b) \in \mathbb{R} \times \mathbb{R}^d$ such that

$$F(\omega, t, c, U + y, \Sigma + z) \leq F(\omega, t, c, U, \Sigma) + ay + b'z.$$

The following example reviews the case of Duffie and Epstein (1992) utility, which includes time-additive discounted expected utility, as well as the continuous-time version of Epstein and Zin (1989) utility. Other examples include multiple-prior formulations as in Chen and Epstein (2002), or (via the equivalence of Skiadas (2002)) in Hansen, Sargent, Turmuhambetova, and Williams (2001). The utility process V is *ordinally equivalent* to U if there exists a progressively measurable function $\chi : \Omega \times [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ such that $\chi(\omega, t, \cdot)$ is strictly increasing for every (ω, t) , and $V_t = \chi(t, U_t)$. In this case, V and U represent the same preference ordering over consumption plans.

Example 13 (Duffie-Epstein Aggregator) F is a Duffie-Epstein aggregator if

$$F(\omega, t, c, U, \Sigma) = a(\omega, t, c, U) - \frac{A(\omega, t, U)}{2} \Sigma' \Sigma, \quad (8)$$

for some functions $a : \Omega \times [0, T] \times \mathcal{C} \times \mathbb{R} \rightarrow \mathbb{R}$ and $A : \Omega \times [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$. Duffie and Epstein (1992) show that A measures comparative risk aversion, and that¹, subject to some technical regularity restrictions, if F is a Duffie-Epstein aggregator, then there is an ordinally equivalent version, V , of U that uniquely solves a BSDE of the form

$$dV_t = -f(t, c_t, V_t) dt + \Sigma_t^{V'} dB_t, \quad V_T = f(T, c_T), \quad (9)$$

for some function $f : \Omega \times [0, T] \times \mathcal{C} \times \mathbb{R} \rightarrow \mathbb{R}$ (which is related to preferences for information in Skiadas (1998)). The usual time-additive discounted expected utility case corresponds to f taking the functional form

$$f(\omega, t, c, V) = u(\omega, t, c) - \beta(\omega, t) V. \quad (10)$$

As a matter of general notation, Σ^U denotes the diffusion coefficient of an Ito process U , so $dU = \dots dt + \Sigma^U dB$. For the case of a quantity expressing returns, such as R , the alternative notation $dR = \dots dt + \sigma^R dB$ is used. Similarly, in SS we write $dU/U = \dots dt + \sigma^U dB$.

To formulate the first-order conditions of optimality for recursive utility, we fix a candidate optimal plan (c, ϕ, W) throughout, and we define the processes $U = U(c)$ and

$$\lambda_t = F_c(t, c_t, U_t, \Sigma_t^U). \quad (11)$$

¹More precisely, Duffie and Epstein consider the case in which the aggregator is state and time independent. The same arguments, however, extend readily to the more general specification considered here.

In a time- t formulation of the agent's problem, λ_t is the Lagrange multiplier for the time- t wealth constraint, since it provides the first-order utility increment (per unit of wealth) as a result of slightly changing time- t wealth. Consumption can be expressed in terms of λ by inverting the above equation: $c_t = \mathcal{I}(t, \lambda_t, U_t, \Sigma_t)$, where the function $\mathcal{I} : \Omega \times [0, T] \times (0, \infty) \times \mathbb{R}^{1+d} \rightarrow \mathcal{C}$ is well-defined implicitly through the equation

$$F_c(t, \mathcal{I}(t, \lambda, U, \Sigma), U, \Sigma) = \lambda, \quad \lambda \in (0, \infty).$$

The utility (super)gradient density can be expressed as a stochastically discounted version of λ as follows. Given any processes $(a, b) \in \mathcal{L}_1(\mathbb{R}) \times \mathcal{L}_2(\mathbb{R}^d)$, the stochastic exponential process $\mathcal{E}(a, b)$ is defined by the SDE

$$\frac{d\mathcal{E}_t(a, b)}{\mathcal{E}_t(a, b)} = a_t dt + b'_t dB_t, \quad \mathcal{E}_0(a, b) = 1. \quad (12)$$

The following is a corollary of Proposition 12 of SS:

Proposition 14 *Suppose that $(F_U, F_\Sigma) \in \mathcal{L}_1(\mathbb{R}) \times \mathcal{L}_2(\mathbb{R}^d)$, $\mathcal{E}(F_U, F_\Sigma) \in \mathcal{S}_2$, and*

$$\pi_t = \mathcal{E}_t(F_U, F_\Sigma) \lambda_t \quad (13)$$

defines an element of $\mathcal{H}$. If $(F_U(t), F_\Sigma(t)) \in (\partial_{U, \Sigma} F)(t, c_t, U_t, \Sigma_t^U)$, then π is a utility supergradient density of U_0 at c .

Applying Ito's lemma to equation (13), and last section's conditions for π to be a state-price density, results in the following constrained FBSDE system (as explained in more detail in the proof of Proposition 16).

Condition 15 (First Order Conditions) *The processes $(U, \Sigma^U, \lambda, \sigma^\lambda, \zeta) \in \mathcal{U} \times \mathcal{L}_2(\mathbb{R}^d) \times \mathcal{L}(\mathbb{R}_{++}) \times \mathcal{L}_2(\mathbb{R}^d) \times \mathcal{L}(\mathbb{R})$, the wealth process W , and the trading strategy ϕ , solve the constrained FBSDE:*

$$\begin{aligned} dU_t &= -F(t, \mathcal{I}(t, \lambda_t, U_t, \Sigma_t^U), U_t, \Sigma_t^U) dt + \Sigma_t^{U'} dB_t, \quad U_T = F(T, W_T), \\ \frac{d\lambda_t}{\lambda_t} &= -(r_t + \zeta_t + F_U(t) + \sigma_t^{\lambda'} F_\Sigma(t)) dt + \sigma_t^{\lambda'} dB_t, \quad \lambda_T = F_c(T, W_T), \\ dW_t &= (W_t r_t + \phi'_t \mu_t^R + e_t - \mathcal{I}(t, \lambda_t, U_t, \Sigma_t^U)) dt + \phi'_t \sigma_t^{R'} dB_t, \quad W_0 = w_0, \\ \varepsilon_t &= \mu_t^R + \sigma_t^{R'} (F_\Sigma(t) + \sigma_t^\lambda), \quad (\phi_t, W_t) \in S, \quad \phi'_t \varepsilon_t - W_t \zeta_t = \delta(\varepsilon_t, \zeta_t), \\ &\quad (F_U(t), F_\Sigma(t)) \in (\partial_{U, \Sigma} F)(t, \mathcal{I}(t, \lambda_t, U_t, \Sigma_t^U), U_t, \Sigma_t^U). \end{aligned}$$

The sufficiency argument in this context can be summarized as follows:

Proposition 16 *Suppose Condition 15 holds, and let $c_t = \mathcal{I}(t, \lambda_t, U_t, \Sigma_t)$ and $\pi = \mathcal{E}(F_U, F_\Sigma) \lambda$. Suppose further that $c \in \mathcal{C}$, $\pi W \in \mathcal{S}_1$, and $\pi, \mathcal{E}(F_U, F_\Sigma) \in \mathcal{S}_2$. Then the plan (c, ϕ, W) is optimal, and $U(c) = U$.*

Remark 17 *The first-order conditions for the case of no intermediate consumption are obtained by omitting the consumption argument in the utility and supergradient dynamics. The optimality verification argument for this case is essentially the same as for the case with intermediate consumption.*

The necessity of Condition 15 for optimality, given a smooth aggregator and the conditions of Proposition 8, follows from Propositions 2 and 8. A necessity result for scale-invariant constraints (and $C = (0, \infty)$) is given in SS.

In a Markovian setting, the above FBSDE system can be formulated as a PDE system, following the Ma, Protter, and Yong (1994) approach. The basic idea is illustrated in SS for the case of scale-invariant constraints, and can easily be adapted to this more general setting. Rather than stating the general system, we later return to the PDE characterization in more specific applications.

5 Translation-Invariant Formulation

This section analyzes a translation-invariant formulation, resulting in a simple relationship between the value function and the wealth process that uncouples the first-order conditions. An alternative (non-Markovian) dynamic programming approach for this formulation is given in Appendix A.

In the remainder of this paper we fix a strictly positive bounded process γ and assume utility dynamics of the form

$$dU_t = -G\left(t, \frac{c_t}{\gamma_t} - U_t, \Sigma_t^U\right) dt + \Sigma_t^{U'} dB_t, \quad U_T = \frac{c_T}{\gamma_T}, \quad (14)$$

where G will be referred to as an *absolute aggregator* (to be contrasted to a proportional aggregator defined in SS). More formally, we henceforth assume:

Condition 18 (Standing Assumption) *The aggregator F takes the form*

$$F(\omega, t, c, U, \Sigma) = G\left(\omega, t, \frac{c}{\gamma_t(\omega)} - U, \Sigma\right), \quad F(\omega, T, c) = \frac{c}{\gamma_T(\omega)},$$

for a function $G : \Omega \times [0, T] \times \mathbb{R}^{1+d} \rightarrow \mathbb{R}$, and a strictly positive bounded process γ . Moreover, the consumption set $\mathcal{C}$ satisfies: $c + x \in \mathcal{C}$ for any $c \in \mathcal{C}$ and any bounded $x \in \mathcal{L}(\mathbb{R})$.

To understand the preference restrictions that lead to the above aggregator form, we introduce some definitions. The utility process U is *translation-invariant relative to γ* , or γ -TI for short, if for any $k \in \mathbb{R}$, $c^1, c^2 \in \mathcal{C}$, and $t \in [0, T]$,

$$U_t(c^1) \geq U_t(c^2) \iff U_t(c^1 + k\gamma) \geq U_t(c^2 + k\gamma).$$

The utility process U is in *certainty equivalent form relative to γ* , or γ -CE for short, if at any time- t information node, $U_t(c)$ represents the units of the

cash flow γ that result in the same time- t utility as the consumption plan c . More formally, U is a γ -CE if for any consumption plan c and any time t , $U_t(c) = U_t(\alpha\gamma)$, where the process α satisfies $\alpha_s = U_t(c)$ for all $s \geq t$, and (say) $\alpha_s = 0$ for $s < t$. In this sense, the cash flow γ is the yardstick relative to which utility is cardinalized. If U is both γ -TI and γ -CE, then it must be quasi-linear relative to γ :

$$U_t(c + k\gamma) = U_t(c) + k \quad \text{for all } k \in \mathbb{R}, c \in \mathcal{C}, \quad (15)$$

a condition that is essentially equivalent to Condition 18.

This claim is isomorphic to corresponding conclusions in SS with regard to homotheticity, via the transformations $\tilde{c}_t = \exp(-c_t/\gamma_t)$ and $\tilde{U}_t(\tilde{c}) = \exp(-U_t(c))$. We note that U is γ -TI if and only if $\tilde{U}$ is homothetic,² and $\tilde{U}$ is 1-CE if and only if U is γ -CE. In SS it is shown that if $\tilde{U}$ is homothetic and 1-CE then $\tilde{U}$ is homogeneous of degree one, a condition that is equivalent to the quasi-linearity property (15) of U . In SS it is also shown that homogeneity of $\tilde{U}$ of degree one corresponds to the dynamics:

$$d\tilde{U}_t/\tilde{U}_t = -\tilde{G}\left(t, \tilde{c}_t/\tilde{U}_t, \tilde{\sigma}_t\right) dt + \tilde{\sigma}_t^{U'} dB_t, \quad \tilde{U}_t = \tilde{c}_T.$$

Letting $G(\omega, t, x, y) = -\tilde{G}(\omega, t, \exp(-x), -y) - y'y/2$, and using Ito's lemma, it follows that the above dynamics are the same as in (14).

Given the above mapping between homothetic and γ -TI formulations, the homothetic utility examples in SS can be mechanically transformed to corresponding γ -TI utility examples. (This does not imply, however, that optimal solutions can also be mechanically transformed. The linearity of the budget constraint is not invariant under the above consumption transformation.) The case of γ -TI Duffie-Epstein utility is reviewed in the following section, including the special case of time-additive exponential utility.

The last assumption of the wealth-invariant formulation is that the agent can trade with no restrictions in the money market and a portfolio ϱ that generates the dividend stream γ (of Condition 18) and has corresponding time- t value Γ_t . More formally, we henceforth assume:

Condition 19 (Standing Assumption) *There exists $\varrho \in \mathbb{R}^m$ and a strictly positive process Γ such that*

$$d\Gamma_t = (r_t\Gamma_t - \gamma_t) dt + \varrho' dR_t, \quad \Gamma_T = \gamma_T,$$

and the constraint set takes the form $S = \{(v, w) : v \in \Phi\}$, for some set Φ such that $v \in \Phi$ implies $v + \alpha\varrho \in \Phi$ for all $\alpha \in \mathbb{R}$.

Given the above condition, $\delta(\varepsilon, \zeta) < \infty$ implies $\zeta = \varrho'\varepsilon = 0$. The first-order conditions therefore imply that the agent correctly prices both the money

²The utility process $\tilde{U}$ is *homothetic* if for any time t , and any $c^1, c^2 \in \mathcal{C}$, and $k \in (0, \infty)$, $\tilde{U}_t(c^1) = \tilde{U}_t(c^2)$ implies $\tilde{U}_t(kc^1) = \tilde{U}_t(kc^2)$.

market asset and the risk-premium of the portfolio ϱ . This intuitive fact will be shown to be a general consequence of optimality. Given the wealth-invariance of the constraint set, we use the simplified notation

$$\delta(\varepsilon) = \delta(\varepsilon, 0) = \sup \{v'\varepsilon : v \in \Phi\}.$$

The key to simplifying the solution is the following informal argument. Given an optimal plan, suppose that, on some time- t event, α units of account are added to the agent's financial wealth. Because of translation invariance with respect to γ , the agent will find it optimal to invest all additional α units of wealth in α/Γ_t shares of the portfolio ϱ , and to consume all of the ensuing dividend stream $\alpha\gamma/\Gamma_t$. Given the quasi-linearity of the utility function with respect to γ , this suggests the following key relationships at the optimum:

$$\lambda_t = \frac{1}{\Gamma_t}, \quad U_t = \frac{1}{\Gamma_t} (Y_t + W_t), \quad \text{and} \quad \phi_t = \phi_t^0 + U_t \varrho, \quad (16)$$

where (Y, ϕ^0) solves a constrained BSDE that is independent of financial wealth.

To state the BSDE, we introduce some notation. The (partial) derivative of $G(\omega, t, \cdot, \Sigma)$ at x is denoted $G_x(\omega, t, x, \Sigma)$, and the superdifferential of the function $G(\omega, t, x, \cdot)$ at Σ is denoted $\partial_\Sigma G(\omega, t, x, \Sigma)$. The functions $\mathcal{X}, G^* : \Omega \times [0, T] \times (0, \infty) \times \mathbb{R}^d \rightarrow \mathbb{R}$ are defined by

$$\begin{aligned} G_x(\omega, t, \mathcal{X}(\omega, t, y, \Sigma), \Sigma) &= y, \\ G^*(\omega, t, y, \Sigma) &= \sup_{x \in \mathbb{R}} \{G(\omega, t, x, \Sigma) - yx\} \\ &= G(\omega, t, \mathcal{X}(\omega, t, y, \Sigma), \Sigma) - y\mathcal{X}(\omega, t, y, \Sigma). \end{aligned}$$

Given the conjectured conditions (16), the first-order conditions reduce to

Condition 20 $(Y, \phi^0) \in \mathcal{L}(\mathbb{R}) \times \mathcal{L}(\mathbb{R}^m)$ satisfies the constrained BSDE:

$$\begin{aligned} dY_t &= \left(Y_t r_t - \phi_t^{0'} \mu_t^R - e_t - \Gamma_t G^* \left(t, \frac{\gamma_t}{\Gamma_t}, \frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \right) + (\sigma_t^R \varrho)' \left(\frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \right) \right) dt \\ &\quad + \Sigma^{Y'} dB, \quad Y_T = e_T, \\ \varepsilon_t &= \mu_t^R + \sigma_t^{R'} \left(G_\Sigma(t) - \frac{1}{\Gamma_t} \sigma_t^R \varrho \right), \quad \phi_t^0 \in \Phi, \quad \delta(\varepsilon_t) = \phi_t^{0'} \varepsilon_t, \\ G_\Sigma(t) &\in \partial_\Sigma G \left(t, \mathcal{X} \left(t, \frac{\gamma_t}{\Gamma_t}, \frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \right), \frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \right). \end{aligned}$$

Given (Y, ϕ^0) and equations (16), the optimal consumption plan is given by

$$c_t = \gamma_t (U_t + x_t), \quad \text{where} \quad x_t = \mathcal{X} \left(t, \frac{\gamma_t}{\Gamma_t}, \frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \right). \quad (17)$$

The wealth process dynamics are obtained by substituting the optimal consumption and trading plan expressions in the budget equation:

$$dW_t = \left(W_t r_t + \frac{Y_t + W_t}{\Gamma_t} (\varrho' \mu_t^R - \gamma_t) + \phi_t^{0'} \mu_t^R + e_t - \gamma_t x_t \right) dt \quad (18)$$

$$+ \left(\phi_t^0 + \frac{Y_t + W_t}{\Gamma_t} \varrho \right)' \sigma_t^{R'} dB_t, \quad W_0 = w_0.$$

The optimality characterization for the wealth-invariant formulation is completed in the following result, proved in Appendix B. (Regarding the assumption of a bounded r in part (b), Remark 9 applies.)

Theorem 21 *Suppose Conditions 18 and 19 hold.*

(a) *(Sufficiency) Suppose Condition 20 is satisfied, the consumption plan c is defined by equation (17), the wealth process W solves SDE (18), and the utility process $U \in \mathcal{U}$ and trading plan ϕ are given by equations (16). Finally, suppose $\mathcal{E} = \mathcal{E}(-\gamma/\Gamma, G_\Sigma) \in \mathcal{S}_2$, as well as $\mathcal{E}/\Gamma \in \mathcal{S}_2$ and $\mathcal{E}W/\Gamma \in \mathcal{S}_1$. Then the plan (c, ϕ, W) is optimal and $U = U(c)$.*

(b) *(Necessity) Suppose the plan (c, ϕ, W) is optimal, $G(\omega, t, x, \Sigma)$ is differentiable in (x, Σ) , $\mathcal{C}$ is a convex cone, and r is bounded. Let $U = U(c)$, $x = c/\gamma - U$, $G_x(t) = G_x(t, x_t, \Sigma_t^U)$, and $G_\Sigma(t) = G_\Sigma(t, x_t, \Sigma_t^U)$, and suppose that $\mathcal{E} = \mathcal{E}(-G_x, G_\Sigma) \in \mathcal{S}_2$ and $1/\Gamma \in \mathcal{S}_2$. Then Condition 20 holds with $Y = U\Gamma - W$ and $\phi^0 = \phi - U\varrho$.*

6 Quasi-Quadratic Aggregator

In this section we further specialize the wealth-invariant formulation of the last section by assuming a tractable quasi-quadratic aggregator form, that includes as special cases all translation-invariant Duffie-Epstein utilities (including time-additive exponential utility), but also allows for first or second-order risk aversion that can depend on the source of uncertainty.

In the remainder of this paper we assume the following condition, using the notation

$$|x|' = (|x_1|, \dots, |x_d|), \quad x \in \mathbb{R}^d.$$

Condition 22 (Standing Assumption) *The absolute aggregator G is quasi-quadratic, meaning that it takes the form*

$$G(\omega, t, x, \Sigma) = g(\omega, t, x) - q(\omega, t)' \Sigma - \kappa(\omega, t)' |\Sigma| - \frac{1}{2} \Sigma' Q(\omega, t) \Sigma, \quad (19)$$

for some (progressively measurable) functions $g : \Omega \times [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$, $q : \Omega \times [0, T] \rightarrow \mathbb{R}^d$, $\kappa : \Omega \times [0, T] \rightarrow \mathbb{R}_+^d$, and $Q : \Omega \times [0, T] \rightarrow \mathbb{R}^{d \times d}$, such that $Q(\omega, t)$ is symmetric positive definite for all (ω, t) . Finally, the processes κ, q , and Q are assumed bounded (for simplicity).

Example 23 *If F is a Duffie and Epstein (1992) aggregator, then the absolute aggregator G must take the quasi-quadratic form $G(\omega, t, x, \Sigma) = g(\omega, t, x) - A(\omega, t) \Sigma' \Sigma / 2$, where increasing A makes the utility more risk averse, without affecting the agent's preferences over deterministic consumption plans. In the special case in which $A(\omega, t) = 1$ and $g(\omega, t, x, y) = \beta(\omega, t) - \exp(-x)$, for some bounded process β , the ordinally equivalent utility $V_t = -\exp(-U_t)$ is time-additive discounted exponential utility:*

$$V_t = E_t \left[\int_t^T -\exp \left(-\int_t^s \beta_u du - \frac{c_s}{\gamma_s} \right) ds - \exp \left(-\int_t^T \beta_u du - \frac{c_T}{\gamma_T} \right) \right].$$

The interpretation of the various terms of equation (19) parallels those of the proportional quasi-quadratic representation in SS. The function g entirely determines the agent's preferences over deterministic consumption plans. The linear term $q' \Sigma$ can be thought of as reflecting the agent's beliefs, in the sense of a Girsanov change of measure (as explained in SS). The remaining coefficients, κ and Q , measure, respectively, first and second-order absolute risk aversion that can be time and state dependent. First or second-order risk aversion can be dependent on the source of risk, a dependence that can be thought of as reflecting the ambiguity of the risk source, or uncertainty in the sense of Ellsberg's experiments.

As in SS, the source-dependence of risk aversion implies that the quasi-quadratic representation depends on the choice of a Brownian motion rotation. More specifically, suppose that $\Theta \in \mathcal{L}(\mathbb{R}^{d \times d})$ satisfies $\Theta'_t \Theta_t = I_d$ (the identity matrix in $\mathbb{R}^d$), and define $\bar{B}_t = \int_0^t \Theta'_s dB_s$, which is a standard Brownian motion generating the same information as B . Defining $\bar{\sigma}_t^R = \Theta'_t \sigma_t^R$, $\bar{\Sigma}_t^U = \Theta'_t \Sigma_t^U$, $\bar{q}_t = \Theta'_t q_t$, and $\bar{Q}_t = \Theta'_t Q_t \Theta_t$, we can write the excess return dynamics as $dR_t = \mu_t^R dt + \bar{\sigma}_t^{R'} d\bar{B}_t$, and the utility dynamics as

$$dU_t = - \left(g \left(t, \frac{c_t}{\gamma_t} - U_t \right) - \bar{q}_t' \bar{\Sigma}_t^U - \kappa_t' |\Theta_t \bar{\Sigma}_t^U| - \frac{1}{2} \bar{\Sigma}_t^{U'} \bar{Q}_t \bar{\Sigma}_t^U \right) dt + \bar{\Sigma}_t^{U'} d\bar{B}_t.$$

The rotation Θ_t can always be chosen so that $\bar{Q}_t$ is diagonal, with the (positive) eigenvalues of Q_t on the diagonal indicating the coefficients of second-order absolute risk aversion to sources of risk as measured by the components of $\bar{\Sigma}^U$. On the other hand, the (also positive) elements of κ measure coefficients of first-order risk aversion to sources of risk as measured by $\Theta \bar{\Sigma}^U$. The Duffie-Epstein utility case corresponds to $q = \kappa = 0$ and $Q = \gamma I$ for some process γ , in which case $\bar{q} = q$ and $\bar{Q} = Q$. The Duffie-Epstein specification is, therefore, invariant to Brownian motion rotations.

With quasi-quadratic utility, we obtain the simplifications

$$\begin{aligned} \mathcal{X}(\omega, t, y, \Sigma) &= \mathcal{X}^g(\omega, t, y), \quad \text{and} \\ G^*(\omega, t, y, \Sigma) &= g^*(\omega, t, y) - q(\omega, t)' \Sigma - \kappa(\omega, t)' |\Sigma| - \frac{1}{2} \Sigma' Q(\omega, t) \Sigma, \end{aligned}$$

where the functions $\mathcal{X}^g, g^* : \Omega \times [0, T] \times (0, \infty) \rightarrow \mathbb{R}$ are defined analogously to $\mathcal{X}$ and G^* by

$$g_x(\omega, t, \mathcal{X}^g(\omega, t, y)) = y, \quad \text{and} \\ g^*(\omega, t, y) = \sup_{x \in \mathbb{R}} (g(\omega, t, x) - yx) = g(\omega, t, \mathcal{X}^g(\omega, t, y)) - y\mathcal{X}^g(\omega, t, y).$$

As in SS, the superdifferential $\partial_\Sigma G$ is characterized as follows. For any $\Sigma \in \mathcal{L}_2(\mathbb{R}^d)$, let $\partial|\Sigma|$ denote the set of all processes Λ that are valued in $[-1, 1]^d$, and whose i^{th} coordinate, Λ_i , satisfies

$$\Lambda_i(t) = 1 \text{ on } \{\Sigma_i(t) > 0\}, \quad \Lambda_i(t) = -1 \text{ on } \{\Sigma_i(t) < 0\}, \\ \text{and } \Lambda_i(t) \in [-1, 1] \text{ on } \{\Sigma_i(t) = 0\}.$$

Moreover, for every $x, y \in \mathbb{R}^d$, $x \otimes y \in \mathbb{R}^d$ denotes element-by-element multiplication, that is, $(x \otimes y)_i = x_i y_i$ for all i . One can easily show:

Lemma 24 $G_\Sigma \in (\partial_\Sigma G)(t)$ if and only if $G_\Sigma(t) = -(q_t + \kappa_t \otimes \Lambda_t + Q_t \Sigma_t^U)$ for some $\Lambda \in \partial|\Sigma^U|$.

Direct computation using the above lemma gives the constrained BSDE of the first-order conditions as in the following proposition. The term $\kappa \otimes \Lambda$ in the first-order conditions is computed explicitly in the following section assuming that Q is diagonal.

Proposition 25 For the quasi-quadratic absolute aggregator G , the constrained BSDE of Condition 20 is equivalent to:

$$dY_t = \left(Y_t r_t - e_t - \Gamma_t g^* \left(t, \frac{\gamma_t}{\Gamma_t} \right) - \delta(\varepsilon_t) + \frac{1}{2\Gamma_t} (\Sigma_t^{Y'} Q_t \Sigma_t^Y - \phi_t^{0'} \sigma_t^{R'} Q_t \sigma_t^R \phi_t^0) \right) dt \\ + \Sigma_t^{Y'} \left[dB_t + \left(q_t + \kappa_t \otimes \Lambda_t + \frac{\sigma_t^R \varrho}{\Gamma_t} \right) dt \right], \quad Y_T = e_T, \\ \phi_t^0 = \Gamma_t (\sigma_t^{R'} Q_t \sigma_t^R)^{-1} \left(\mu_t^R - \varepsilon_t - \sigma_t^{R'} \left[q_t + \kappa_t \otimes \Lambda_t + \frac{\sigma_t^R \varrho + Q_t \Sigma_t^Y}{\Gamma_t} \right] \right), \\ \phi_t^0 \in \Phi, \quad \delta(\varepsilon_t) = \phi_t^{0'} \varepsilon_t, \quad \Lambda_t \in \partial|\Sigma_t^Y + \sigma_t^R \phi_t^0|.$$

Remark 26 If $\Phi = \mathbb{R}^m$, $m \leq d$, corresponding to incomplete ($m < d$) or complete markets ($m = d$), the first-order conditions simplify by letting $\varepsilon = 0$. Another simplification arises if the processes $(r, \gamma, \mu^R, \sigma^R, \kappa, q, Q)$ are deterministic, and g is state-independent, in which case the above first-order conditions apply with $\Sigma^Y = 0$.

One can now restate Theorem 21 for the quasi-quadratic case by replacing Condition 20 with the above constrained BSDE. Given a solution to the latter, the complete solution is recovered using equations (16), (17), and (18).

Example 27 Suppose that the quasi-quadratic absolute aggregator G is smooth (that is, $\kappa = 0$). As in Example 33 of SS, a simple expression for the optimal trading strategy is obtained under a single linear constraint. Suppose $\Phi = \{v \in \mathbf{R}^m : \alpha \leq l'v \leq \beta\}$, where $l \in \mathbf{R}^m$ and $-\infty \leq \alpha \leq \beta \leq \infty$. For example, a short-selling constraint on asset i corresponds to $\alpha = 0$, $\beta = \infty$, and l a vector with a one in the i^{th} position and zeros elsewhere. Defining

$$\phi_t^{0*} = A_t \left(\mu_t^R - \sigma_t^{R'} \left[q_t + \frac{\sigma_t^R \varrho + Q_t \Sigma_t^Y}{\Gamma_t} \right] \right), \quad A_t = \Gamma_t (\sigma_t^{R'} Q_t \sigma_t^R)^{-1},$$

the pair (ϕ^0, ε) in the first-order conditions of Proposition 25 is given by

$$\phi_t^0 = \phi_t^{0*} - A_t \varepsilon_t, \quad \varepsilon_t = -(l' A_t l)^{-1} l (\min \{ \max \{ l' \phi_t^{0*}, \alpha \}, \beta \} - l' \phi_t^{0*}).$$

Note that ϕ^{0*} is what the optimal expression for ϕ^0 , as a function of Σ^Y , would be in the unconstrained case. The above claim follows easily from the first-order conditions. The details, as well as the extension to multiple linear constraints, are analogous to Example 33 in SS, and are therefore omitted.

7 Solutions under Quasi-Quadratic Aggregator

The remainder of the main part of this paper deals with more detailed solutions under a quasi-quadratic absolute aggregator, including more explicit representations of the effect of first-order risk aversion, and of the effect of an undiversifiable income stream. All of this section's analysis is non-Markovian. Markovian applications are presented in the final section.

The following condition will be henceforth assumed:

Condition 28 (Standing Assumption) The excess return volatility process, σ^R , and the second-order risk aversion process, Q , take the form

$$\sigma^R = \begin{bmatrix} \sigma_{MM}^R \\ 0 \end{bmatrix} \quad \text{and} \quad Q = \begin{bmatrix} Q_{MM} & 0 \\ 0 & Q_{NN} \end{bmatrix},$$

where σ_{MM}^R and Q_{MM} are processes valued in $\mathbb{R}^{m \times m}$.

Remark 29 Strictly speaking the above condition entails no loss in generality, since one can always set $m = d$ and model missing markets through the constraint set Φ (see Remark 1 of SS). But the more interesting applications will assume $m < d$. By Lemma 10 of SS, σ^R can always be put in the above form by passing to a new Brownian motion that generates the same filtration as the original one. If $Q = \alpha I$ for some $\alpha \in \mathbb{R}$ and the identity matrix I (as is the case with Duffie-Epstein utility, including time-additive utility) the change of Brownian motion does not affect Q , and therefore imposing Condition 28 entails no loss in generality (even if $m < d$).

The notational scheme suggested in the above condition will be followed throughout: M and N refer to the sets of *marketed* and *non-marketed* securities, respectively, and are also used as indices of corresponding matrix blocks. In particular, given any d -dimensional process or vector x , we write $x' = [x'_M, x'_N]$, where x_M is valued in $\mathbb{R}^m$ and x_N is valued in $\mathbb{R}^{d-m}$. Given the new subscripts, we often omit dummy time indices to simplify the notation. For example, the above condition implies that $dR = \mu^R dt + \sigma_{MM}^{R'} dB_M$. It is important to note that, unless otherwise stated, the coefficients μ^R and σ_{MM}^R are not required to be adapted to the filtration generated by B_M .

This section's main topic is the simplification of the first-order conditions of Proposition 25. For this purpose, the following processes will be of repeated use:

$$\eta_M = (\sigma_{MM}^{R'})^{-1} (\mu^R - \varepsilon) \quad \text{and} \quad \bar{\eta}_M = \eta_M - q_M - \frac{1}{\Gamma} \sigma_{MM}^R \varrho. \quad (20)$$

Moreover, in the first order conditions, we will often refer to the term

$$\hat{\eta}_M = \bar{\eta}_M - \kappa_M \otimes \Lambda_M. \quad (21)$$

For example, in this context, the optimal trading strategy is $\phi = \phi^0 + U\varrho$, where

$$\phi^0 = (\sigma_{MM}^R)^{-1} (\Gamma Q_{MM}^{-1} \hat{\eta}_M - \Sigma_M^Y). \quad (22)$$

Example 30 (Fictitious Completion) *Suppose that $m < d$ and $\Phi \subseteq \mathbb{R}^m$, meaning that markets are incomplete, and trading in the available assets can be further restricted. We can embed this market into a fictitious complete market by introducing $d - m$ fictitious securities in which trading is unconstrained, and with price dynamics specified by $\mu_N^R = q_N + \kappa_N \otimes \Lambda_N + \Gamma^{-1} Q_{NN} \Sigma_N^Y$, σ_{NN}^R equal to the identity matrix, and $\sigma_{MN}^R = \sigma_{NM}^{R'} = 0$. In the first-order conditions of optimality in the fictitious market, ε_M is the same as in the original market, while $\varepsilon_N = 0$. Moreover, the optimal trading strategy as given in Proposition 16 implies that $\phi_N = 0$. That is, we have selected μ_N^R so that the impossibility of trading in fictitious assets is a non-binding constraint. Direct computation shows that the BSDE for Y is the same in the fictitious market and the original market. Finally, equation (3) with $\zeta = 0$ and $\eta' = [\eta'_M, \mu'_N]$ specifies the state-price density (up to positive scaling) of the fictitious complete market discussed at the end of Section 3.*

The following result clarifies the role of first-order risk aversion in the first-order conditions, given a diagonal process Q .

Proposition 31 *Suppose Condition 28 holds, Q is diagonal, and the process $\bar{\eta}_M$ is defined in (20). Then, in the first-order conditions of Proposition 25, the terms $\kappa \otimes \Lambda$ and ϕ^0 can equivalently be specified by*

$$\kappa_M \otimes \Lambda_M = \max \{ \min \{ \bar{\eta}_M, \kappa_M \}, -\kappa_M \}, \quad \kappa_N \otimes \Lambda_N = \kappa_N \otimes \text{sign}(\Sigma_N^Y),$$

and equation (22), where $\hat{\eta}_M$ can be written (equivalently to (21)) as

$$\hat{\eta}_M = \min \{ \max \{ 0, \bar{\eta}_M - \kappa_M \}, \bar{\eta}_M + \kappa_M \} = Q_{MM} \Sigma_M^U.$$

Given the assumption that Q is diagonal, the last equation implies that, for any $i \in M$, $\Sigma_i^U = 0$ whenever $|\bar{\eta}_i| \leq \kappa_i$; that is, the agent completely hedges the i^{th} dimension of risk, as a consequence of first-order risk aversion. Analogous results for the scale-invariant case are presented in SS (see also Segal and Spivak (1990), Chen and Epstein (2002), and Epstein and Miao (2000)).

Example 32 Suppose further that the parameters $r, e, \gamma, \mu^R, \sigma^R, \kappa, q, Q$ are all deterministic, and the function g is state independent. Then Y is also deterministic, and therefore $\Sigma^Y = 0$ and $\phi^0 = (\sigma_{MM}^R)^{-1} \Gamma \Sigma_M^U$. In this case, $\Sigma_i^U = 0$ can imply complete nonparticipation in the market for asset i . For example, suppose that σ_{MM}^R (as well as Q) is diagonal, and $\Phi = \mathbf{R}^m$. Suppose also that the portfolio ϱ consists of a share in a asset one, which is a consol bond generating dividend stream γ . Then, for any $i \geq 2$, $|(\mu_i^R / \sigma_{ii}^R) - q_i| \leq \kappa_i$ implies $\phi_i = 0$. That is, a sufficiently small belief-adjusted market price of the risk traded by asset $i \geq 2$ results in the agent's non-participation in this market.

This section's last main result focuses on the role of a non-traded endowment income stream. We assume that $\Phi = \mathbb{R}^m$, which implies that markets can be incomplete ($m < d$), but there are no further trading constraints. We also assume that all parameters, with the exception of the endowment income stream e , depend only on uncertainty generated by B_M , and we derive a decomposition of Y into a market and an endowment component.

Proposition 33 In addition to Condition 22, suppose that $\Phi = \mathbb{R}^m$, and the processes $r, \gamma, \mu^R, \sigma^R, \kappa_M, q_M, Q_{MM}$, and $g^*(\gamma/\Gamma)$ are all adapted to the filtration, $\{\mathcal{F}_t^M\}$, generated by B_M . Then the process Y of the first-order conditions, and its volatility, can be decomposed as

$$Y = Y^M + Y^e, \quad \Sigma_M^Y = \Sigma_M^{Y^M} + \Sigma_M^{Y^e}, \quad \text{and} \quad \Sigma_N^Y = \Sigma_N^{Y^e},$$

where $(Y^M, \Sigma_M^{Y^M})$ is adapted to $\{\mathcal{F}_t^M\}$ and solves the BSDE:

$$\begin{aligned} dY^M = & - \left(\Gamma g^* \left(\frac{\gamma}{\Gamma} \right) - Y^M r + \frac{\Gamma}{2} \hat{\eta}_M' Q_{MM}^{-1} \hat{\eta}_M \right) dt \\ & + \Sigma_M^{Y^M'} (dB_M + \eta_M dt), \quad Y_T^M = 0, \end{aligned}$$

while (Y^e, Σ^{Y^e}) is adapted to the finer filtration $\{\mathcal{F}_t\}$ and solves the BSDE:

$$\begin{aligned} dY^e = & - \left(e - Y^e r - \frac{1}{2\Gamma} \Sigma_N^{Y^e'} Q_{NN} \Sigma_N^{Y^e} \right) dt \\ & + \Sigma_M^{Y^e'} (dB_M + \eta_M dt) + \Sigma_N^{Y^e'} (dB_N + (q_N + \kappa_N \otimes \Lambda_N) dt), \quad Y_T^e = e_T. \end{aligned}$$

To present integrated versions of the above BSDE's, we introduce some notation with regard to Girsanov changes of measure. Given any $b \in \mathcal{L}(\mathbb{R}^d)$, we use the notation B^b and ξ^b , to denote the processes satisfying

$$dB_t^b = dB_t + b_t dt, \quad B_0^b = 0, \quad \text{and} \quad \frac{d\xi_t^b}{\xi_t^b} = -b_t' dB_t, \quad \xi_0^b = 1. \quad (23)$$

If ξ^b is a martingale (for example, if b satisfies the Novikov condition), then B^b is Brownian motion under the probability P^b with density $dP^b/dP = \xi_T^b$. The expectation operator corresponding to the probability P^b is denoted E^b .

Consider now the processes

$$\theta = \begin{bmatrix} \eta_M \\ q_N + \kappa_N \otimes \Lambda_N \end{bmatrix} \quad \text{and} \quad \mathcal{B}_{t,s} = \exp \left(- \int_t^s r_u du \right).$$

and assume that ξ^θ is a martingale. The above BSDE's can then be integrated (provided the integrals are well-defined), to obtain the following expressions for the two components of Y :

$$\begin{aligned} Y_t^M &= E^\theta \left[\int_t^T \mathcal{B}_{t,s} \Gamma_s \left(g^* \left(s, \frac{\gamma_s}{\Gamma_s} \right) + \frac{1}{2} \hat{\eta}'_M(s) Q_{MM}^{-1}(s) \hat{\eta}_M(s) \right) ds \mid \mathcal{F}_t^M \right] \\ Y_t^e &= E^\theta \left[\int_t^T \mathcal{B}_{t,s} \left(e_s - \frac{1}{2\Gamma_s} \Sigma_N^{e'}(s) Q_{NN}(s) \Sigma_N^e(s) \right) ds + \mathcal{B}_{t,T} e_T \mid \mathcal{F}_t \right]. \end{aligned}$$

Example 34 (Complete Markets) Suppose that markets are complete, that is, $m = d$ and $\Phi = \mathbb{R}^m$. Then $\eta = (\sigma^{R'})^{-1} \mu^R$ is the unique market price of risk process, and $Y_t^e = E_t^\eta \left[\int_t^T \mathcal{B}_{t,s} e_s ds + \mathcal{B}_{t,T} e_T \right]$ is the time- t market value of the endowment.

Example 35 We present two special cases of Proposition 33 in which Y^e is given by an expression of the form

$$\exp \left(- \frac{a}{\Gamma_t} Y_t^e \right) = E_t^b \left[\exp \left(- \int_t^T \frac{a}{\Gamma_s} e_s ds - \frac{a}{\Gamma_T} e_T \right) \right], \quad (24)$$

for properly defined a and b , extending a more special result by Musiela and Zariphopoulou (2001). The proof of the following claims is given in Appendix B. In addition to the assumptions of Proposition 33, we assume that there is no intermediate consumption (meaning that for every $c \in \mathcal{C}$, $c_t = 0$ for $t < T$), and in particular $\gamma_t = 0$ for all $t < T$ (which presents no problems since the strict positivity of γ_T implies that of Γ). Even though we have not treated the case of no-intermediate consumption separately, the results are essentially the same (see Remark 17). We also assume that r and $\gamma_T > 0$ are deterministic, and that $\kappa_N = 0$. Within this setting we consider two special cases:

Case 1 (idiosyncratic endowment). Suppose that e is adapted to the filtration generated by B_N , and $Q_{NN}(t) = aI$, where $a > 0$ and I is an identity matrix. Then $\Sigma_M^{Y^M} = 0$ and equation 24 holds for any b such that $b_N = q_N$, provided ξ^b is a martingale and the stated integral exists.

Case 2 (non-idiosyncratic endowment). Suppose that Q_{NN} is constant over time; η_M and q_N are deterministic; and the endowment process satisfies

$$de_t = \alpha_t dt + \beta_t \Sigma' dB_t,$$

where $\Sigma \in \mathbb{R}^d$, and α and β are real-valued and adapted to the filtration generated by e . Then equation 24 holds with $a = \Sigma'_N Q_{NN} \Sigma_N / \Sigma' \Sigma$ and $b' = \theta' = [\eta'_M, q'_N]$. The solution of Musiela and Zariphopoulou (2001) is the subcase obtained by assuming $Q = I$ (time-additive utility), $q = 0$, Markovian dynamics for e and R , and a single lognormally distributed risky asset.

8 Markovian Incomplete-Market Applications

We conclude the main part of this paper with an outline of Markovian applications with incomplete markets, starting with a PDE version of the BSDE of the first-order conditions, and followed by a particularly tractable class of problems in which Y becomes a quadratic function of the Markov state, with deterministic coefficients that solve a system of ordinary differential equations (of the Riccati form). The underlying Markov process and price dynamics are similar to those in Liu (2001), extending those considered in the last section of Schroder and Skiadas (1999). The analysis parallels the last section in SS for the scale-invariant case.

In addition to Condition 28, throughout this section, we assume that the aggregator is smooth (that is, $\kappa = 0$), and $\Phi = \mathbb{R}^m$. Markets can therefore be incomplete ($m < d$), but there are no other constraints, implying that $\varepsilon = 0$ in the first-order conditions. For simplicity (and without loss in generality), we also assume that $q = 0$. In particular, expressions (20) simplify to

$$\eta_M = (\sigma_{MM}^{R'})^{-1} \mu^R \quad \text{and} \quad \bar{\eta}_M = \eta_M - \frac{1}{\Gamma} \sigma_{MM}^R \varrho.$$

Uncertainty is driven by a k -dimensional Markov process, Z , solving the SDE

$$dZ_t = \mu^Z(t, Z_t)dt + \sigma^Z(t, Z_t)' dB_t.$$

The price dynamics parameters are assumed to be possibly time-dependent but deterministic functions of Z . With some convenient duplication of notation, we assume the Markovian functional relationships:

$$\begin{aligned} r_t &= r(t, Z_t), \quad \mu_t^R = \mu^R(t, Z_t), \quad \sigma_t^R = \sigma^R(t, Z_t), \\ e_t &= e(t, Z_t), \quad \gamma_t = \gamma(t, Z_t), \quad \Gamma_t = \Gamma(t, Z_t). \end{aligned} \quad (25)$$

We also write $g^*(t, \gamma_t/\Gamma_t) = g^*(t, Z_t)$. All the functions of the state Z are assumed to be at least twice continuously differentiable.

Letting $Y = Y(t, Z_t)$, the PDE corresponding to the BSDE of Proposition 25 is

$$\begin{aligned} Y_t + Y'_Z \mu_t^Z + \frac{1}{2} \text{tr} (Y_{ZZ} \sigma^{Z'} \sigma^Z) &= Yr - e - \Gamma g^* + Y'_Z \sigma_M^{Z'} \eta_M \\ &\quad + \frac{1}{2\Gamma} Y'_Z \sigma_N^{Z'} Q_{NN} \sigma_N^Z Y_Z - \frac{\Gamma}{2} \bar{\eta}'_M Q_{MM}^{-1} \bar{\eta}_M, \end{aligned}$$

with terminal condition $Y(T, \cdot) = e(T, \cdot)$. Here Y_t, Y_Z , and Y_{ZZ} denote partial derivatives of the function Y . (In particular, Y_t should not be confused with the time- t value of the process Y . While there is duplication in notation, the meaning should be clear from the context.) Expanding $Y(t, Z)$ by Ito's lemma (given that Y is sufficiently smooth), a standard argument shows that (under appropriate integrability assumptions) a solution to the above PDE leads to a corresponding solution of the BSDE by (Y, Σ^Y) , where $\Sigma_t^Y = \sigma^Z(t, Z_t) Y_Z(t, Z_t)$. We omit the standard details, and instead identify formally a class of problems for which the above PDE reduces to an ODE system.

We partition the components of Z into two parts: $Z_t = [Z_t^{a'}, Z_t^{b'}]'$, with $Z_t^a \in \mathbb{R}^a$ and $Z_t^b \in \mathbb{R}^b$, treating a and b both as indices of the blocks of Z , as well as integers indicating the respective dimensionality of Z^a and Z^b (and therefore $a + b = k$). The dynamics of Z are assumed to take the form

$$d \begin{pmatrix} Z_t^a \\ Z_t^b \end{pmatrix} = \mu^Z(Z_t) dt + \begin{pmatrix} \sigma^{Za'} \\ \sigma^{Zb}(Z_t)' \end{pmatrix} dB_t,$$

where $\mu^Z : \mathbb{R}^k \rightarrow \mathbb{R}^k$, $\sigma^{Zb} : \mathbb{R}^k \rightarrow \mathbb{R}^{d \times b}$, and $\sigma^{Za} \in \mathbb{R}^{d \times a}$. Further restrictions on the functions μ^Z and σ^{Zb} are imposed below.

The idea is to back-engineer conditions that lead to a solution of the form

$$Y(t, z) = Y_0(t) + Y_1'(t) z + \frac{1}{2} z^{a'} Y_2(t) z^a, \quad (26)$$

for deterministic coefficients (Y_0, Y_1, Y_2) , and then use the PDE for Y to derive an ODE system for (Y_0, Y_1, Y_2) .

A set of assumptions that make the above plan work is presented below. We assume, without loss of generality, that all the matrices appearing in quadratic forms, including Y_2 above, are symmetric.

Assumption 1 The processes r and γ are deterministic (which implies that Γ is also deterministic and $\varrho = 0$).

Assumption 2 The functions g^* and $\eta'_M Q_{MM}^{-1} \eta_M$ are linear in Z_t and quadratic in Z_t^a :

$$\Gamma(t, z) g^*(t, z) + \frac{1}{2} \Gamma(t, z) \eta_M(t, z)' Q_{MM}^{-1}(t, z) \eta_M(t, z) = p_0 + p_1' z + \frac{1}{2} z^{a'} p_2 z^a,$$

where $p_0 \in \mathbb{R}$, $p_1 \in \mathbb{R}^k$, $p_2 \in \mathbb{R}^{a \times a}$. (For example, the assumption on g^* is satisfied if $g(\omega, t, x) = \beta(t, Z(\omega, t)) + \bar{g}(t, x)$, where β is linear in Z and quadratic in Z^a , and $\bar{g}$ is state-independent.)

Assumption 3 The endowment e_t is linear in Z_t and quadratic in Z_t^a :

$$e(t, z) = h_0 + h_1' z + \frac{1}{2} z^{a'} h_2 z^a,$$

where $h_0 \in \mathbb{R}$, $h_1 \in \mathbb{R}^k$, $h_2 \in \mathbb{R}^{a \times a}$.

Assumption 4 The $R^{k \times k}$ -valued process $\Gamma^{-1} \sigma_N^{Z'} Q_{NN} \sigma_N^Z$ has the quadratic structure:

$$\begin{aligned} \Gamma(t, z)^{-1} \sigma_N^Z(t, z)' Q_{NN}(t, z) \sigma_N^Z(t, z) &= \begin{bmatrix} D_0^{aa} & D_0^{ab} \\ D_0^{ab'} & D_0^{bb} \end{bmatrix} \\ &+ \begin{bmatrix} 0 & [z^{a'} D_1^{ab}[i, j]]_{i=1, \dots, a; j=1, \dots, b} \\ [z^{a'} D_1^{ab}[i, j]]'_{i=1, \dots, a; j=1, \dots, b} & [z' D_1^{bb}[i, j]]_{i, j=1, \dots, b} \end{bmatrix} \\ &+ \begin{bmatrix} 0 & 0 \\ 0 & [z^{a'} D_2^{bb}[i, j] z^a]_{i, j=1, \dots, b} \end{bmatrix}, \end{aligned}$$

where $D_0^{aa} \in \mathbb{R}^{a \times a}$, $D_0^{ab} \in \mathbb{R}^{a \times b}$, $D_0^{bb} \in \mathbb{R}^{b \times b}$, $D_1^{ab}[i, j] \in \mathbb{R}^a$, $D_1^{bb}[i, j] \in \mathbb{R}^k$, and $D_2^{bb}[i, j] \in \mathbb{R}^{a \times a}$. We define D_0 as the above $k \times k$ matrix with quadrants D_0^{ij} , $i, j \in \{a, b\}$.

Assumption 5 The drift of Z_t^a plus $\sigma^{Za'} \eta_M$ is linear in Z_t^a ; and the drift of Z_t^b plus $\sigma^{Zb}(t, z)' \eta_M$ is linear in Z_t and quadratic in Z_t^a :

$$\mu^Z(t, z) - \sigma_M^Z(t, z)' \eta_M(t, z) = L_0 + \begin{bmatrix} L_1^a z^a \\ L_1^b z + [z^{a'} L_2^b[i] z^a]_{i=1, \dots, b} \end{bmatrix},$$

where $L_0 = [L_0^a, L_0^b]' \in \mathbb{R}^{a+b}$, $L_1^a \in \mathbb{R}^{a \times a}$, $L_1^b \in \mathbb{R}^{b \times k}$, and $L_2^b[i] \in \mathbb{R}^{a \times a}$.

Define the $\mathbb{R}^{a \times a}$ -valued process Ξ , whose i^{th} row is

$$\Xi_{i*} = \sum_{j=1}^b Y_{1j}^b D_1^{ab}[i, j]'.$$

Imposing Assumptions 1-5, substituting the conjectured solution (26) into the PDE for Y , and setting the constant, linear and quadratic terms involving Z to zero, we obtain the following system of Riccati equations:

$$\begin{aligned} 0 &= p_0 + h_0 - rY_0 + \dot{Y}_0 + \frac{1}{2} \text{trace}(Y_2 \sigma^{Za'} \sigma^{Za}) - \frac{1}{2} Y_1' D_0 Y_1 + Y_1' L_0 \\ 0 &= p_1 + h_1 + \dot{Y}_1 - rY_1 + L_1^{b'} Y_1^b - \frac{1}{2} \sum_{i=1}^b \sum_{j=1}^b Y_{1i}^b Y_{1j}^b D_1^{bb}[i, j] \\ &\quad + \begin{bmatrix} -Y_2 [D_0^{aa}, D_0^{ab}] Y_1 + Y_2 L_0^a + L_1^{a'} Y_1^a - \sum_{i=1}^a \sum_{j=1}^b Y_{1i}^a Y_{1j}^b D_1^{ab}[i, j] \\ 0 \end{bmatrix} \\ 0 &= p_2 + h_2 + \dot{Y}_2 - rY_2 - Y_2 D_0^{aa} Y_2 + Y_2 L_1^a + L_1^a Y_2 - (Y_2 \Xi + \Xi' Y_2) \\ &\quad - \sum_{i=1}^b \sum_{j=1}^b Y_{1i}^b Y_{1j}^b D_2^{bb}[i, j] + 2 \sum_{i=1}^b Y_{1i}^b L_2^b[i] \end{aligned}$$

with terminal conditions

$$Y_0(T) = h_0(T), \quad Y_1(T) = h_1(T), \quad Y_2(T) = h_2(T).$$

A continuously differentiable solution (Y_0, Y_1, Y_2) to the Riccati system leads to a corresponding solution to the PDE for Y .

Example 36 *In addition to Assumption 1, we assume a diagonal and deterministic second-order risk aversion matrix:*

$$Q_{MM}(t, z) = \alpha_M(t)I_M, \quad Q_{NN}(t) = \alpha_N(t)I_N, \quad t \in [0, T],$$

where α_M and α_N are deterministic real-valued processes. We also assume that g^* is state independent (which implies that $g^*(t, \gamma_t/\Gamma_t)$ is deterministic, since we assume deterministic r and γ above). Assumptions 2-5 above are then satisfied by the two classes of dynamics defined below, and also studied in Schroder and Skiadas (1999) in the context of complete markets and power or logarithmic recursive utility.

Class 1: $Z = Z^a$ ($b = 0$), the state vector and market-price of risk follow Ornstein-Uhlenbeck processes, and the endowment is quadratic in the state vector:

$$\begin{aligned} dZ_t &= (\mu - \alpha Z_t) dt + \Sigma' dB_t, \\ \eta_M &= v + HZ_t, \quad e_t = h_0 + h_1' Z_t + \frac{1}{2} Z_t' h_2 Z_t, \end{aligned}$$

where $\mu \in \mathbb{R}^a$, $\Sigma \in \mathbb{R}^{d \times a}$, $\alpha \in \mathbb{R}^{a \times a}$, $v \in \mathbb{R}^m$, $H \in \mathbb{R}^{m \times a}$, $h_0 \in \mathbb{R}$, $h_1 \in \mathbb{R}^a$, $h_2 \in \mathbb{R}^{a \times a}$. An Ornstein-Uhlenbeck market price of risk process is used in Kim and Omberg (1996), and in the variation of the Stein-Stein model examined by Liu (2002), both in the context of an investor maximizing expected power utility of terminal wealth.

Class 2: $Z = Z^b$ ($a = 0$) and the endowment e , as well as each element of η_M' , μ^Z , $\sigma_N^{Z'} \sigma_N^Z$ and $\sigma_M^{Z'} \eta_M$ are affine in Z . For example, letting $\text{diag}(x)$ denote a diagonal matrix with the vector x as its diagonal, this class includes the specification:

$$\begin{aligned} dZ_t &= (\mu - \alpha Z_t) dt + \Sigma' \text{diag}(\sqrt{v + HZ_t}) dB_t, \\ \eta_M &= \text{diag}(\sqrt{v + HZ_t}) \varphi, \quad e_t = h_0 + h_1' Z_t, \end{aligned}$$

where $\mu \in \mathbb{R}^b$, $\Sigma \in \mathbb{R}^{d \times b}$, $\alpha \in \mathbb{R}^{b \times b}$, $v \in \mathbb{R}^d$, $H \in \mathbb{R}^{d \times b}$, $\varphi \in \mathbb{R}^{d \times m}$, $h_0 \in \mathbb{R}$, $h_1 \in \mathbb{R}^b$. In this case the ODE system can be solved sequentially from the bottom up. This class includes the square-root volatility process of Heston (1993), and the square-root market-price-of-risk process in Chacko and Viceira (1999) (in the context of log-Duffie-Epstein utility with initial wealth only).

A Appendix: Dynamic Programming Approach

In this appendix we provide an alternative dynamic programming verification argument in the wealth invariant setting, complementing the verification argument given in Schroder and Skiadas (2002b) for the scale-invariant case. As in the utility gradient approach, a Markovian structure is not required.

The setting throughout this appendix is that of Section 5. To informally motivate the argument, we let $J_t(w)$ be the agent's time- t optimal utility value given time- t financial wealth w . As in Section 5, we conjecture that $J_t(w) = (Y_t + w)/\Gamma_t$, where Γ is defined in Condition 19, and Y is an Ito process that does not depend on wealth, and whose dynamics are denoted

$$dY_t = \mu_t^Y dt + \Sigma_t^{Y'} dB_t, \quad Y_T = e_T.$$

Using hats to denote (candidate) optimal quantities, suppose that $(\hat{c}, \hat{\phi}, \hat{W})$ is an optimal plan, with corresponding utility process $\hat{U} = U(\hat{c})$, and define

$$\hat{x}_t = \frac{\hat{c}_t}{\gamma_t} - \frac{Y_t + \hat{W}_t}{\Gamma_t} \quad \text{and} \quad \hat{\phi}^0 = \hat{\phi} - \frac{Y_t + \hat{W}_t}{\Gamma_t} \varrho. \quad (27)$$

Optimality requires that $\hat{U}_t = J_t(\hat{W}_t) = (Y_t + \hat{W}_t)/\Gamma_t$. Applying Ito's lemma, using the budget equation and the dynamics for Γ , we conclude that

$$\Gamma_t d\hat{U}_t = \mathcal{D}_t(\hat{x}_t, \hat{\phi}_t^0) dt + (\Sigma_t^Y + \sigma_t^R \hat{\phi}_t^0)' dB_t,$$

where

$$\mathcal{D}_t(x_t, \phi_t^0) = \mu_t^Y - r_t Y_t + \phi_t^{0'} \mu_t^R + e_t - (\sigma_t^R \varrho)' \left(\frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \right) - \gamma_t x_t.$$

Comparing to the dynamics of $\hat{U}$, we conclude that at the optimum

$$\mathcal{D}_t(\hat{x}_t, \hat{\phi}_t^0) + \Gamma_t G\left(t, \hat{x}_t, \frac{\Sigma_t^Y + \sigma_t^R \hat{\phi}_t^0}{\Gamma_t}\right) = 0. \quad (28)$$

Given any other feasible plan (c, ϕ, W) , and letting

$$x_t = \frac{c_t}{\gamma_t} - \frac{Y_t + W_t}{\Gamma_t} \quad \text{and} \quad \phi^0 = \phi - \frac{Y_t + W_t}{\Gamma_t} \varrho, \quad (29)$$

the usual dynamic programming argument suggests the inequality

$$\mathcal{D}_t(x_t, \phi_t^0) + \Gamma_t G\left(t, x_t, \frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t}\right) \leq 0. \quad (30)$$

The theorem below verifies the above conjectures, using the following technical regularity condition:

Condition 37 Given any feasible plan (c, ϕ, W) , the process $J \equiv (Y + W) / \Gamma$ is in $\mathcal{S}_2$, and there exist processes (F_U, F_Σ) such that

$$(F_U(t), F_\Sigma(t)) \in (\partial_{U, \Sigma} F)(c_t, J_t, \Sigma_t^J) \quad \text{and} \quad \mathcal{E}(F_U, F_\Sigma) \in \mathcal{S}_2.$$

Theorem 38 Suppose the utility is given by Conditions 12 and 18, the strictly positive process Γ and the constraint set Φ are given by Condition 19, and Condition 37 holds. If the feasible plan $(\hat{c}, \hat{\phi}, \hat{W})$ satisfies equation (28), where $(\hat{x}, \hat{\phi}^0)$ is defined in (27), while any other feasible plan (c, ϕ, W) satisfies inequality (30), where (x, ϕ^0) is defined in (29), then $(\hat{c}, \hat{\phi}, \hat{W})$ is optimal.

Proof. Consider any feasible plan (c, ϕ, W) , and let $U = U(c) \in \mathcal{U} \subseteq \mathcal{S}_2$ and $J = (Y + W) / \Gamma \in \mathcal{S}_2$. We will show that $U_0 \leq J_0$, with equality holding if $(c, \phi, W) = (\hat{c}, \hat{\phi}, \hat{W})$.

Applying Ito's lemma to J and using the budget equation (18) and the dynamics of Γ , we find

$$\begin{aligned} dJ_t &= \frac{1}{\Gamma_t} \left(\mu_t^Y - r_t Y_t + \phi_t^{0'} \mu_t^R + e_t - (\sigma_t^R \varrho)' \Sigma_t^J - \gamma_t x_t \right) dt + \Sigma_t^{J'} dB_t \\ \Sigma_t^J &= \frac{1}{\Gamma_t} (\Sigma_t^Y + \sigma_t^R \phi_t^0). \end{aligned}$$

Let p^1 be the nonnegative (by inequality (30)) process such that

$$0 = \frac{1}{\Gamma_t} \left(\mu_t^Y - r_t Y_t + \phi_t^{0'} \mu_t^R + e_t - (\sigma_t^R \varrho)' \Sigma_t^J - \gamma_t x_t \right) + G(t, x_t, \Sigma_t^J) + p_t^1.$$

Then the above dynamics for J can be written as

$$dJ_t = - (F(t, c_t, J_t, \Sigma_t^J) + p_t^1) dt + \Sigma_t^{J'} dB_t.$$

On the other hand, $dU_t = -F(t, c_t, U_t, \Sigma_t^U) dt + \Sigma_t^{U'} dB_t$. Letting $y = U - J$, and $z = \Sigma^U - \Sigma^J$, we therefore have

$$dy_t = - (F(t, c_t, U_t, \Sigma_t^U) - F(t, c_t, J_t, \Sigma_t^J) - p_t^1) dt + z_t' dB_t$$

Selecting $(F_U, F_\Sigma) \in (\partial_{U, \Sigma} F)(c, J, \Sigma^J)$ as in Condition 37,

$$F(t, c_t, U_t, \Sigma_t^U) - F(t, c_t, J_t, \Sigma_t^J) = F_U(t) y_t + F_\Sigma(t) z_t - p_t^2,$$

for some nonnegative process p^2 . Therefore, with $p = p^1 + p^2$,

$$dy_t = - (F_U(t) y_t + F_\Sigma(t)' z_t - p_t) dt + z_t' dB_t.$$

Letting $\mathcal{E} = \mathcal{E}(F_U, F_\Sigma) \in \mathcal{S}_2$, integration by parts gives

$$d(\mathcal{E}_t y_t) = \mathcal{E}_t dy_t + y_t d\mathcal{E}_t + dy_t d\mathcal{E}_t = \mathcal{E}_t p_t dt + dM_t,$$

for a local martingale M . Let $\{\tau_N\}$ be a corresponding localizing stopping time sequence converging to T almost surely. Then

$$y_0 = E \left[- \int_0^{\tau_N} \mathcal{E}_s p_s ds + \mathcal{E}_{\tau_N} y_{\tau_N} \right] \leq E [\mathcal{E}_{\tau_N} y_{\tau_N}].$$

The last term converges since $\mathcal{E}y \in \mathcal{S}_1$, and therefore $y_0 \leq E[y_T \mathcal{E}_T] = 0$ (since $y_T = 0$). We have proved that $U_0 \leq J_0$.

Repeating the argument for $(c, \phi^0, W) = (\hat{c}, \hat{\phi}^0, \hat{W})$ and $p^1 = 0$, gives $U_0 = J_0$, proving optimality of $(\hat{c}, \hat{\phi}, \hat{W})$, and completing the proof. ■

Conditions (28) and (30) are clearly implied by the Bellman equation

$$0 = \max_{x \in \mathbb{R}, \phi^0 \in \Phi} \mathcal{D}_t(x, \phi^0) + \Gamma_t G \left(t, x, \frac{\Sigma_t^Y + \sigma_t^R \phi^0}{\Gamma_t} \right), \quad (31)$$

with the optimal $\hat{x}_t$ and $\hat{\phi}_t^0$ providing the maximizing values. We conclude by relating the Bellman equation to the first-order conditions of optimality as stated in Condition 20. For the sake of brevity, we assume tht $G(\omega, t, \cdot)$ is smooth. (The analogous argument applies using the supergradient of G .)

Maximization with respect to x is equivalent to

$$\hat{x}_t = \mathcal{X} \left(t, \frac{\gamma_t}{\Gamma_t}, \frac{\Sigma_t^Y + \sigma_t^R \hat{\phi}_t^0}{\Gamma_t} \right).$$

Substituting this expression into equation (28) shows that μ_t^Y matches the drift of Y in Condition 20.

The gradient of the function maximized in (31) with respect to ϕ^0 at $\hat{\phi}^0$ is

$$\varepsilon_t = \mu_t^R + \sigma_t^{R'} \left(G_\Sigma \left(t, \hat{x}_t, \frac{\Sigma_t^Y + \sigma_t^R \hat{\phi}_t^0}{\Gamma_t} \right) - \frac{1}{\Gamma_t} \sigma_t^R \varrho \right).$$

Maximization over $\phi^0 \in \Phi$ implies that $(\phi^0 - \hat{\phi}^0)' \varepsilon \leq 0$ for every $\phi^0 \in \Phi$, a condition that is equivalent to $\hat{\phi}^{0'} \varepsilon = \delta(\varepsilon)$.

Based on the above observations, one can now easily confirm that the first-order conditions of Condition 20 imply the Bellman equation (31), and therefore the above verification argument applies.

B Appendix: Proofs

This appendix contains proofs omitted from the main text.

B.1 Proof of Proposition 6

Given any feasible plan $(\tilde{c}, \tilde{\phi}, \tilde{W})$ and stopping time τ , integration by parts, the dynamics of π , and the budget equation imply

$$\begin{aligned} \pi_\tau \tilde{W}_\tau - \pi_0 w_0 &= \int_0^\tau \pi_t \left(\tilde{\phi}'_t \varepsilon_t - \tilde{W}_t \zeta_t \right) dt - \int_0^\tau \pi_t (\tilde{c}_t - e_t) dt \\ &\quad + \int_0^\tau \pi_t \left(\sigma_t^R \tilde{\phi}_t - \tilde{W}_t \eta_t \right)' dB_t, \end{aligned} \quad (32)$$

and therefore

$$\pi_\tau \tilde{W}_\tau - \pi_0 w_0 \leq \int_0^\tau \pi_t \delta(\varepsilon_t, \zeta_t) dt - \int_0^\tau \pi_t (\tilde{c}_t - e_t) dt + M_t^1,$$

for a local martingale M^1 . Given Condition 3, the same argument applied at (c, ϕ, W) gives

$$\pi_\tau W_\tau - \pi_0 w_0 = \int_0^\tau \pi_t \delta(\varepsilon_t, \zeta_t) dt - \int_0^\tau \pi_t (c_t - e_t) dt + M_t^2,$$

for another local martingale M^2 . Letting $x = \tilde{c} - c$, and $M = M^1 - M^2$, we conclude that

$$\pi_\tau \tilde{W}_\tau - \pi_\tau W_\tau \leq - \int_0^\tau \pi_t x_t dt + M_\tau.$$

Let $\{\tau_n : n = 1, 2, \dots\}$ be an increasing sequence of stopping times that converges to T and such that M stopped at τ_n is a martingale. Taking expectations in the last inequality, we find

$$0 \geq E \left[\int_0^{\tau_n} \pi_t x_t dt + \pi_{\tau_n} \tilde{W}_{\tau_n} - \pi_{\tau_n} W_{\tau_n} \right]. \quad (33)$$

The idea is to let $n \rightarrow \infty$ to conclude that $(\pi|x) \leq 0$. There remains to justify the interchange of limits and integration. For the integral of πx , we apply the dominated convergence theorem, using the assumptions $\pi, x \in \mathcal{H}$. For the term, involving $\pi \tilde{W}$, we use the assumptions $\pi \in \mathcal{S}_2$ and $\tilde{W}^- \in \mathcal{S}_2$, and Fatou's lemma to conclude that

$$\liminf_{t \rightarrow \infty} E \left[\pi_{\tau_n} \tilde{W}_{\tau_n} \right] \geq E \left[\pi_T \tilde{W}_T \right].$$

Finally, the assumption $\pi W \in \mathcal{S}_1$ implies that $E[\pi_{\tau_n} W_{\tau_n}]$ converges to $E[\pi_T W_T]$. Taking the limit inferior on both sides of inequality (33), we therefore reach the conclusion $(\pi|x) \leq 0$.

B.2 Proof of Lemma 7

As in the last proof, equation (32) holds for any feasible plan $(\tilde{c}, \tilde{\phi}, \tilde{W})$ and stopping time τ . Taking the difference of the equation applied at $(c + x, \phi + y, W + z)$, and the equation applied at (c, ϕ, W) , we find that

$$\int_0^\tau \pi_t x_t dt + \pi_\tau z_\tau = \int_0^\tau \pi_t \Delta_t dt + M_\tau,$$

for some local martingale M . Consider now an increasing sequence of stopping times $\{\tau_n : n = 1, 2, \dots\}$ that converges to T , such that M stopped at τ_n is a martingale. It then follows that

$$E \left[\int_0^{\tau_n} \pi_t x_t dt + \pi_{\tau_n} z_{\tau_n} \right] = E \left[\int_0^{\tau_n} \pi_t \Delta_t dt \right], \quad n = 1, 2, \dots$$

Feasibility of $(c + x, \phi + y, W + z)$ implies that $x_T = z_T$, and therefore the left-hand side converges to $(\pi|x)$, given the assumptions that $\pi z \in \mathcal{S}_1$ and $\pi, x \in \mathcal{H}$ and $\pi z \in \mathcal{S}_1$. The right-hand side converges to $(\pi|\Delta)$ since $\pi, \Delta \in \mathcal{H}$.

B.3 Proof of Proposition 8

By taking the accumulated value of a unit invested at time zero in the money market as the numeraire, we assume without loss in generality, since r is assumed bounded, that $r = 0$.

Step 1: Proof that $\zeta = 0$. Consider any time $t < T$ and any event $F \in \mathcal{F}_t$. Given any $h \in (t, T - t)$, consider the feasible plans $(c + x, \theta, W + z)$ and $(c - x, \theta, W - z)$, where x is equal to the indicator function of $F \times [t, t + h]$, while z is given by $dz_s = -x_s ds$, $z_0 = 0$. Since π is a state-price density, it follows that $(\pi|x) = 0$, and therefore

$$E \left[1_F \left(\frac{1}{h} \int_t^{t+h} \pi_s ds - \pi_T \right) \right] = 0.$$

Since π is continuous and $\pi \in \mathcal{S}_1$, taking the limit implies that $E[1_F \pi_t] = E[1_F \pi_T]$. Applying this condition over all t and $F \in \mathcal{F}_t$, implies that π is a martingale, and therefore $\zeta = 0$.

Step 2: Proof that $\phi'_t \varepsilon_t = \delta(\varepsilon_t, 0)$. We show that, for every process $\phi + y$ valued in Φ , $y'_t \varepsilon_t \leq 0$. Suppose instead that there exists some process y such that $\tilde{\phi} = \phi + y$ is valued in Φ but $E \left[\int_0^T 1 \{y'_t \varepsilon_t > 0\} dt \right] > 0$. For each positive integer N , define

$$y_t^N = y_t 1 \{0 \leq y'_t \varepsilon_t \leq N, \|y_t\| < N\}.$$

Since $1\{y_t^{N'}\varepsilon_t > 0\}$ monotonically converges almost surely to $1\{y_t'\varepsilon_t > 0\}$, it follows that $E\left[\int_0^T 1\{y_t^{N'}\varepsilon_t > 0\} dt\right] > 0$ for some N . We therefore assume, without loss in generality, that $y'\varepsilon$ is non-negative and bounded, and y is bounded, which in turn implies that $\tilde{\phi}$ is a trading plan. Similarly, for every positive integer N , define the stopping time

$$\tau^N = \inf \left\{ t : \int_0^t y'_s dR_s > N \quad \text{or} \quad t = T \right\},$$

and the process $y_t^N = y_t 1\{t < \tau^N\}$. The same convergence argument used above shows that $E\left[\int_0^T 1\{y_t^{N'}\varepsilon_t > 0\} dt\right] > 0$ for some N . We therefore also assume, without loss in generality, that the process z , defined by $dz_t = y'_t dR_t$, $z_0 = 0$, is bounded, and therefore $W + z$ is a wealth process. Letting $x_t = 0$ for $t < T$, and $x_T = z_T$, it follows that $(c + x, \phi + y, W + z)$ is a feasible plan. Applying Lemma 7, it follows that $(\pi|x) = (\pi|y'\varepsilon) > 0$, implying that π cannot be a state-price density.

B.4 Proof of Proposition 16

Letting $d\lambda/\lambda = \mu^\lambda dt + \sigma^{\lambda'} dB$, equation (13) implies

$$\frac{d\pi_t}{\pi_t} = (F_U(t) + \mu_t^\lambda + \sigma_t^{\lambda'} F_\Sigma(t)) dt + (F_\Sigma(t) + \sigma_t^\lambda)' dB_t.$$

Comparing to equation 3, we obtain

$$\mu_t^\lambda = -(r_t + \zeta_t + F_U(t) + \sigma_t^{\lambda'} F_\Sigma(t)) \quad \text{and} \quad \sigma_t^\lambda = -(F_\Sigma(t) + \eta_t).$$

Using these expressions along with Condition 3, the trading plan constraint, the utility dynamics, and the budget equation, results in the claimed set of first-order conditions as a constrained FBSDE system. Given these observations, the result follows easily from Propositions 2, 6, and 14.

B.5 Proof of Theorem 21

We begin with some basic consequences of the wealth-invariant formulation. The aggregator form implies that

$$F_c(t, c_t, U_t, \Sigma_t) = \frac{1}{\gamma_t} G_x\left(t, \frac{c_t}{\gamma_t} - U_t, \Sigma_t\right),$$

and therefore

$$c_t = \mathcal{I}(t, \lambda_t, U_t, \Sigma_t) \iff \frac{c_t}{\gamma_t} - U_t = \mathcal{X}(t, \lambda_t \gamma_t, \Sigma_t). \quad (34)$$

The aggregator form and the last observation implies that $(F_U(t), F_\Sigma(t)) \in (\partial_{U, \Sigma} F)(t, \mathcal{I}(t, \lambda_t, U_t, \Sigma_t), U_t, \Sigma_t)$ if and only if

$$F_U(t) = -G_x(t, \mathcal{X}(t, \lambda_t \gamma_t, \Sigma_t), \Sigma_t) \quad \text{and} \quad F_\Sigma(t) \in \partial_\Sigma G(t, \mathcal{X}(t, \lambda_t \gamma_t, \Sigma_t), \Sigma_t).$$

We also recall that $\phi = \phi^0 + U\varrho \in \Phi$ if and only if $\phi^0 \in \Phi$, and that $\delta(\varepsilon_t, \zeta_t) < \infty$ implies $\zeta_t = \varrho'_t \varepsilon_t = 0$ and $\delta(\varepsilon_t, \zeta_t) = \delta(\varepsilon_t)$. Using these observations, we now show the sufficiency and necessity of the proposed first-order conditions.

(a) (Sufficiency) We confirm that the proposed solution satisfies the conditions of Proposition 16. Given the construction of U and ϕ in (16), and equation (17), the wealth dynamics (18) are easily confirmed to be equivalent to the budget equation (1). If $dU_t = \dots dt + \Sigma_t^{U'} dB_t$, integration by parts applied to $Y = U\Gamma - W$ implies

$$\Sigma_t^U = \frac{\Sigma_t^Y + \sigma_t^R \phi_t^0}{\Gamma_t} \quad \text{and} \quad x_t = \frac{c_t}{\gamma_t} - U_t = \mathcal{X}\left(t, \frac{\gamma_t}{\Gamma_t}, \Sigma_t^U\right). \quad (35)$$

By the equivalence (34) and equation (17), $c_t = \mathcal{I}(t, \lambda_t, U_t, \Sigma_t)$, resulting in the wealth dynamics of Condition 15.

Since $\lambda = 1/\Gamma$, if $d\lambda_t/\lambda_t = \mu_t^\lambda dt + \sigma_t^{\lambda'} dB_t$, then

$$\mu_t^\lambda = -r_t + \frac{\gamma_t}{\Gamma_t} - \frac{\varrho'_t \mu_t^R}{\Gamma_t} + \sigma_t^{\lambda'} \sigma_t^\lambda \quad \text{and} \quad \sigma_t^\lambda = -\frac{\sigma_t^R \varrho}{\Gamma_t}.$$

The expression for σ^λ implies that the last two lines of Condition 15 can be obtained from the conditions:

$$\varepsilon_t = \mu_t^R + \sigma_t^{R'} \left(G_\Sigma(t) - \frac{\sigma_t^R \varrho}{\Gamma_t} \right), \quad \phi_t^0 \in \Phi, \quad \delta(\varepsilon_t) = \phi_t^{0'} \varepsilon_t, \quad \zeta_t = 0,$$

by letting $F_\Sigma(t) = G_\Sigma(t) \in \partial_\Sigma G(t, x_t, \Sigma_t)$. The expression for μ^λ implies the dynamics of λ in Condition 15 with $F_U(t) = -G_x(t, x_t, \Sigma_t)$, $F_\Sigma(t) = G_\Sigma(t)$, and $\zeta = 0$. To see that, we use $x_t = \mathcal{X}(t, \gamma_t/\Gamma_t, \Sigma_t)$, $\varrho'_t \varepsilon_t = 0$, and $\Gamma_t \sigma_t^\lambda = -\sigma_t^R \varrho$, to conclude that

$$\frac{\gamma_t}{\Gamma_t} = G_x(t, x_t, \Sigma_t) + \frac{\varrho'_t \varepsilon_t}{\Gamma_t} = -F_U(t) + \frac{\varrho'_t \mu_t^R}{\Gamma_t} - \sigma_t^{\lambda'} (F_\Sigma(t) + \sigma_t^\lambda).$$

Substituting in the expression for μ^λ above we obtain the drift term of λ in Condition 15 (with $\zeta = 0$).

The last part of Condition 15 left to verify are the dynamics of U . For this purpose, we use the above derived expression for Σ^U and the fact that

$$G^*\left(t, \frac{\gamma_t}{\Gamma_t}, \Sigma\right) = G(t, x_t, \Sigma) - \frac{\gamma_t}{\Gamma_t} x_t$$

to rewrite the dynamics of Y as

$$dY_t = \left(Y_t r_t - \phi_t^{0'} \mu_t^R - e_t - \Gamma_t G(t, x_t, \Sigma_t^U) + \gamma_t x_t + (\sigma_t^R \varrho)' \Sigma_t^U \right) dt + \Sigma_t^{Y'} dB_t.$$

Integration by parts applied to $Y = \Gamma U - W$ gives $dY_t = U_t d\Gamma_t + \Gamma_t dU_t + d\Gamma'_t dU_t - dW_t$, or equivalently

$$dU_t = \frac{1}{\Gamma_t} \left(dY_t - U_t d\Gamma_t + dW_t - (\sigma_t^R \varrho)' \Sigma_t^U dt \right).$$

Substituting in the above the dynamics for Y , Γ , and W , and simplifying using the identities (16) and (17), we obtain $dU_t = -G(t, x_t, \Sigma_t^U) dt + \Sigma_t^{U'} dB_t$, which, given the aggregator specification, are the same utility dynamics as in Condition 15.

Finally, optimality follows by Proposition 16 with $\pi_t = \mathcal{E}_t(F_U, F_\Sigma) / \Gamma_t$ and $F_U(t) = -\gamma_t / \Gamma_t$, using the assumption that γ is bounded.

(b) (Necessity) We begin by proving that $\lambda = 1/\Gamma$. Since G is smooth, by Proposition 14, $\pi \in \mathcal{E}\lambda$, where $\lambda_t = G_x(t, x_t, \Sigma_t) / \gamma_t$, defines a utility gradient density. By Proposition 2, π is also a state-price density. Since trading the portfolio ϱ is unrestricted,

$$\Gamma_t = \frac{1}{\pi_t} E_t \left[\int_t^T \pi_s \gamma_s ds + \pi_T \gamma_T \right].$$

This follows by a standard argument that we now outline. Suppose, without loss of generality (by taking one unit invested initially in the money market as the numeraire), that $r = 0$. We have shown in Proposition 8 that π is a martingale. Suppose now that at time t and event $F \in \mathcal{F}_t$ the agent borrows from the money market to buy portfolio ϱ , consuming the ensuing dividend stream γ , and paying back the loan Γ_t by reducing terminal consumption. The resulting feasible incremental cash flow x is given by $x_s = 0$ for $s < t$, $x_s = 1_F \gamma_s$ for $s \in (t, T)$, and $x_T = 1_F (\gamma_T - \Gamma_t)$. Since $-x$ is also feasible, we have

$$0 = (\pi|x) = E \left[1_F \left(\int_t^T \pi_s \gamma_s ds + \pi_T (\gamma_T - \Gamma_t) \right) \right].$$

Since $F \in \mathcal{F}_t$ is arbitrary, it follows that $E_t \left[\int_t^T \pi_s \gamma_s ds + \pi_T \gamma_T \right] = \Gamma_t E_t [\pi_T] = \Gamma_t \pi_t$, as claimed.

Define $f(k) = U(c + k\gamma)$. Taking the derivative with respect to k at zero, we obtain:

$$\mathcal{E}_t f'_t(0) = E_t \left(\int_t^T \pi_s \gamma_s ds + \pi_s \gamma_T \right) = \pi_t \Gamma_t.$$

Since $U(c + k\gamma) = k + U(c)$, it is also the case that $f'(0) = 1$, and therefore $\Gamma_t = \mathcal{E}_t / \pi_t = 1/\lambda_t$.

Having shown the key fact that $\lambda = 1/\Gamma$, we define the process $Y = U\Gamma - W$, and apply Ito's lemma, using the dynamics of U and Γ , the definition of G^* and $\mathcal{X}$, as well as the budget equation, to obtain

$$\begin{aligned} dY_t &= U_t d\Gamma_t + \Gamma_t dU_t + \Sigma_t^{U'} (\sigma^R \varrho) dt - dW_t \\ &= \left(Y_t r_t - \phi_t^0 \mu_t^R + \Sigma_t^{U'} \sigma_t^R \varrho - \Gamma_t G^* \left(t, \frac{\gamma_t}{\Gamma_t}, \Sigma_t^U \right) - e_t \right) dt \\ &\quad + (\Gamma_t \Sigma_t^U - \sigma^R \phi_t^0)' dB_t \end{aligned}$$

The diffusion term confirms the expression for Σ^U in (35), which when substituted into the drift term results in the dynamics of Y in Condition 20. The rest of the condition follows from the smoothness of G , and Propositions 2 and 8, which imply the validity of Condition 3 with $\zeta = 0$.

B.6 Proof of Proposition 25

Substituting

$$G_\Sigma(t) = -(q_t + \kappa_t \otimes \Lambda_t + Q_t \Sigma^U), \quad \Lambda_t \in \partial |\Sigma^U|,$$

into the equation for ε_t in Condition 20 gives

$$\varepsilon_t = \mu_t^R - \sigma_t^{R'} \left(q_t + \kappa_t \otimes \Lambda_t + \frac{\sigma_t^R \varrho}{\Gamma_t} + Q_t \Sigma_t^U \right).$$

The formula for ϕ_t^0 is obtained after substituting $\Sigma^U = \Gamma_t^{-1} (\Sigma_t^Y + \sigma_t^R \phi_t^0)$. Computing the inner product of ε_t and ϕ_t^0 results in

$$\phi_t^{0'} \mu_t^R = \delta(\varepsilon_t) + \phi_t^{0'} \sigma_t^{R'} (q_t + \kappa_t \otimes \Lambda_t + \Gamma_t^{-1} \sigma_t^R \varrho + Q_t \Sigma_t^U).$$

Letting $dY_t = \mu_t^Y dt + \Sigma_t^{Y'} dB_t$, we substitute for G_t^* in Condition 20 to find

$$\mu_t^Y = Y_t r_t - \phi_t^{0'} \mu_t^R - e_t - \Gamma_t g^*(t) + (\sigma_t^R \varrho)' \Sigma_t^U + \Gamma_t \left(q_t' \Sigma_t^U + \kappa_t' |\Sigma_t^U| + \frac{1}{2} \Sigma_t^{U'} Q_t \Sigma_t^U \right).$$

Using the fact that $\kappa(t)' |\Sigma_t^U| = \Sigma_t^{U'} (\kappa_t \otimes \Lambda_t)$ and the above expression for $\phi_t^{0'} \mu_t^R$, the drift term of Y becomes

$$\begin{aligned} \mu_t^Y &= Y_t r_t - e_t - \Gamma_t g^*(t) - \delta(\varepsilon_t) + (\Gamma_t \Sigma_t^{U'} - \phi_t^{0'} \sigma_t^{R'}) (q_t + \kappa_t \otimes \Lambda_t + \Gamma_t^{-1} \sigma_t^R \varrho) \\ &\quad + \frac{1}{2} \Gamma_t \Sigma_t^{U'} Q_t \Sigma_t^U - \phi_t^{0'} \sigma_t^{R'} Q_t \Sigma_t^U \\ &= Y_t r_t - e_t - \Gamma_t g^*(t) - \delta(\varepsilon_t) + \Sigma_t^{Y'} (q_t + \kappa_t \otimes \Lambda_t + \Gamma_t^{-1} \sigma_t^R \varrho) \\ &\quad + \frac{1}{2} \Gamma_t^{-1} (\Sigma_t^{Y'} Q_t \Sigma_t^Y - \phi_t^{0'} \sigma_t^{R'} Q_t \sigma_t^R \phi_t^0). \end{aligned}$$

B.7 Proof of Proposition 31

From the expression for ϕ^0 in (22), we obtain

$$\Gamma^{-1} Q_{ii} (\Sigma^Y + \sigma^R \phi^0)_i = \bar{\eta}_M - \kappa_i \Lambda_i, \quad i \in M.$$

Since $\Lambda_i \in \partial |\Sigma^Y + \sigma^R \phi^0|$, there are three possibilities: (1) $(\Sigma^Y + \sigma^R \phi^0)_i > 0$, $\Lambda_i = 1$, $\bar{\eta}_i > \kappa_i$; (2) $(\Sigma^Y + \sigma^R \phi^0)_i = 0$, $\bar{\eta}_i = \kappa_i \Lambda_i$; and (3) $(\Sigma^Y + \sigma^R \phi^0)_i < 0$, $\Lambda_i = -1$, $\bar{\eta}_i < -\kappa_i$. In all three cases, $\kappa_M \otimes \Lambda_M = \max \{ \min \{ \bar{\eta}_M, \kappa_M \}, -\kappa_M \}$. Since $(\Sigma^Y + \sigma^R \phi^0)_N = \Sigma_N^Y$, we have $\Lambda_N \in \partial |\Sigma_N^Y|$ and therefore $\Sigma_N^{Y'} (\kappa_N \otimes \Lambda_N) = \Sigma_N^{Y'} (\kappa_N \otimes \text{sign}(\Sigma_N^Y))$. Finally, from $\Gamma U = Y + W$, we get $\Gamma \Sigma_M^U = \Sigma_M^Y + \sigma_{MM}^R \phi^0 = \Gamma (Q_{MM})^{-1} \hat{\eta}_M$.

B.8 Proof of Proposition 33

Substituting the expression for ϕ^0 in (22) into the equation for Y in Proposition 25, we find

$$\begin{aligned} dY &= \left(Yr - e - \Gamma g^* \left(\frac{\gamma}{\Gamma} \right) - \frac{\Gamma}{2} \hat{\eta}' Q_{MM}^{-1} \hat{\eta} + \frac{1}{2\Gamma} \Sigma_N^{Y'} Q_{NN} \Sigma_N^Y \right) dt \\ &\quad + \Sigma^{Y'} \left[dB + \left(\frac{\eta_M}{q_N + \kappa_N \otimes \Lambda_N} \right) dt \right]. \end{aligned}$$

Suppose Y_t^e satisfies the stated BSDE and $\Sigma_N^M = 0$. Then direct computation shows that $Y_t^M = Y - Y_t^e$ satisfies the claimed BSDE.

B.9 Proof of Example 35

Only the proof Case 2 is presented. The proof of Case 1 is similar. The assumptions of the example imply that e is the only source of uncertainty driving Y_t^e , and therefore $\Sigma_t^{Y^e} = \nu_t \Sigma$ for some scalar-valued process ν . Therefore,

$$dY_t^e = - \left(e_t - Y_t^e r_t - \frac{a}{2\Gamma_t} \nu_t^2 \Sigma' \Sigma \right) dt + \nu_t \Sigma' dB_t^\theta.$$

Defining $z_t = \exp(-aY_t^e/\Gamma_t)$, we have

$$\begin{aligned} \frac{dz_t}{z_t} &= -\frac{a}{\Gamma_t} dY_t^e + \frac{1}{2} \left(\frac{a}{\Gamma_t} \right)^2 \nu_t^2 \Sigma' \Sigma dt - Y_t^e d \left(\frac{a}{\Gamma_t} \right) \\ &= \frac{a}{\Gamma_t} (e_t - Y_t^e r_t) dt - \frac{a}{\Gamma_t} \nu_t \Sigma' dB_t^\theta - Y_t^e d \left(\frac{a}{\Gamma_t} \right) \\ &= -aY_t^e \left(d \left(\frac{1}{\Gamma_t} \right) + \frac{r_t}{\Gamma_t} dt \right) + \frac{a}{\Gamma_t} e_t dt + \sigma_t^{z'} dB_t^\theta, \end{aligned}$$

where $\sigma_t^z = -a\nu_t \Sigma^e / \Gamma_t$. Since r and γ are deterministic,

$$d \left(\frac{1}{\Gamma_t} \right) = - \left(\frac{r_t - \gamma_t / \Gamma_t}{\Gamma_t} \right) dt,$$

and therefore z solves the BSDE:

$$\frac{dz_t}{z_t} = \frac{1}{\Gamma_t} (ae_t - \gamma_t \log(z_t)) dt + \sigma_t^{z'} dB_t^\theta, \quad z_T = \exp \left(-\frac{a}{\Gamma_T} e_T \right).$$

Under the assumption that $\gamma_t = 0$, $t < T$, the equation for z in (24) is easily confirmed.

References

- ANDERSON, E., L. HANSEN, AND T. SARGENT (2000): “Robustness, Detection and the Price of Risk,” working paper, Dept. of Economics, University of Chicago.
- CHACKO, G., AND L. VICEIRA (1999): “Dynamic Consumption and Portfolio Choice with Stochastic Volatility in Incomplete Markets,” working paper, Harvard University.
- CHEN, Z., AND L. EPSTEIN (2002): “Ambiguity, Risk, and Asset Returns in Continuous Time,” *Econometrica*, 70, 1403–1443.
- COX, J., AND C.-F. HUANG (1989): “Optimal Consumption and Portfolio Policies When Asset Prices Follow a Diffusion Process,” *Journal of Economic Theory*, 49, 33–83.
- CUOCO, D. (1997): “Optimal Consumption and Equilibrium Prices with Portfolio Constraints and Stochastic Income,” *Journal of Economic Theory*, 72, 33–73.
- CVITANIĆ, J., AND I. KARATZAS (1992): “Convex Duality in Constrained Portfolio Optimization,” *The Annals of Applied Probability*, 2, 767–818.
- DETEMPLE, J., AND F. ZAPATERO (1991): “Asset Prices in an Exchange Economy with Habit Formation,” *Econometrica*, 59, 1633–1657.
- DUFFIE, D. (2001): *Dynamic Asset Pricing Theory*. Princeton University Press, Princeton, New Jersey, third edn.
- DUFFIE, D., AND L. EPSTEIN (1992): “Stochastic Differential Utility,” *Econometrica*, 60, 353–394.
- DUFFIE, D., D. FILIPOVIC, AND W. SCHACHERMAYER (2001): “Affine Processes and Applications in Finance,” working paper.
- DUFFIE, D., V. FLEMING, M. SONER, AND T. ZARIPHOUPOULOU (1997): “Hedging in Incomplete Markets with HARA Utility,” *Journal of Economic Dynamics and Control*, 21, 753–782.
- DUFFIE, D., AND R. KAN (1996): “A Yield-Factor Model of Interest Rates,” *Mathematical Finance*, 6, 379–406.
- DUFFIE, D., AND C. SKIADAS (1994): “Continuous-Time Security Pricing: A Utility Gradient Approach,” *Journal of Mathematical Economics*, 23, 107–131.
- EL KAROUI, N., S. PENG, AND M.-C. QUENEZ (2001): “A Dynamic Maximum Principle for the Optimization of Recursive Utilities Under Constraints,” *Annals of Applied Probability*, pp. 664–693.

- EPSTEIN, L., AND J. MIAO (2000): “A Two-Person Dynamic Equilibrium under Ambiguity,” working paper, Dept. of Economics, Univ. of Rochester, Rochester, NY.
- EPSTEIN, L., AND S. ZIN (1989): “Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework,” *Econometrica*, 57, 937–969.
- HANSEN, L., T. SARGENT, G. TURMUHAMBETOVA, AND N. WILLIAMS (2001): “Robustness and Uncertainty Aversion,” working paper, Dept. of Economics, University of Chicago.
- HE, H., AND N. PEARSON (1991): “Consumption and Portfolio Policies with Incomplete Markets and Short-Sale Constrains: The Infinite Dimensional Case,” *Journal of Economic Theory*, 54, 259–304.
- HESTON, S. (1993): “A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options,” *Review of Financial Studies*, 6, 327–344.
- HUGONNIER, J., AND D. KRAMKOV (2002): “Optimal Investment with Random Endowments in Incomplete Markets,” working paper, HEC Montreal and Carnegie Mellon University.
- KARATZAS, I., J. LEHOCZKY, AND S. SHREVE (1987): “Optimal Portfolio and Consumption Decisions for a ‘Small Investor’ on a Finite Horizon,” *SIAM Journal of Control and Optimization*, 25, 1557–1586.
- KARATZAS, I., J. LEHOCZKY, S. SHREVE, AND G. XU (1991): “Martingale and Duality Methods for Utility Maximization in an Incomplete Market,” *SIAM Journal of Control and Optimization*, 29, 702–730.
- KARATZAS, I., AND S. SHREVE (1998): *Methods of Mathematical Finance*. Springer, New York.
- KRAMKOV, D., AND W. SCHACHERMAYER (1999): “The Asymptotic Elasticity of Utility Functions and Optimal Investment in Incomplete Markets,” *Annals of Applied Probability*, 9, 904–950.
- LAZRAK, A., AND M. C. QUENEZ (2002): “A Generalized Stochastic Differential Utility,” working paper, University of British Columbia.
- LIU, J. (2001): “Portfolio Selection in Stochastic Environments,” working paper, UCLA.
- MA, J., P. PROTTER, AND J. YONG (1994): “Solving Forward-Backward Stochastic Differential Equations Explicitly – A Four Step Scheme,” *Probability Theory and Related Fields*, 98, 339–359.
- MAENHOUT, P. (1999): “Robust Portfolio Rules and Asset Pricing,” working paper, INSEAD.

- MERTON, R. (1971): “Optimum Consumption and Portfolio Rules in a Continuous-Time Model,” *Journal of Economic Theory*, 3, 373–413.
- MUSIELA, M., AND T. ZARIPHOPULOU (2001): “Pricing and Risk Management of Derivatives Written on Non-Traded Assets,” working paper.
- PIAZZESI, M. (2002): “Affine Term Structure Models,” working paper.
- SCHRODER, M., AND C. SKIADAS (1999): “Optimal Consumption and Portfolio Selection with Stochastic Differential Utility,” *Journal of Economic Theory*, 89, 68–126.
- (2002a): “An Isomorphism Between Asset Pricing Models with and Without Linear Habit Formation,” *Review of Financial Studies*, 15, 1189–1221.
- SCHRODER, M., AND C. SKIADAS (2002b): “Optimal Lifetime Consumption-Portfolio Strategies under Trading Constraints and Generalized Recursive Preferences,” Working Paper No. 285, Dept. of Finance, Kellogg School of Management, Northwestern University.
- SEGAL, U., AND A. SPIVAK (1990): “First-Order versus Second-Order Risk Aversion,” *Journal of Economic Theory*, 51, 111–125.
- SHREVE, S., AND G. XU (1992): “A Duality Method for Optimal Consumption and Investment under Short-Selling Prohibition: I, General Market Coefficients; and II, Constant Market Coefficients,” *Annals of Applied Probability*, 2, 87–112 and 314–328.
- SKIADAS, C. (1998): “Recursive Utility and Preferences for Information,” *Economic Theory*, 12, 293–312.
- SKIADAS, C. (2002): “Robust Control and Recursive Utility,” *Finance and Stochastics*, forthcoming.
- ZARIPHOPULOU, T. (1994): “Consumption-Investment Models with Constraints,” *SIAM Journal of Control and Optimization*, 32, 59–85.